

Shapley Meets Uniform: An Axiomatic Framework for Attribution in Online Advertising

Raghav Singal

Tuck School of Business, Dartmouth College, singal@dartmouth.edu

Omar Besbes

Graduate School of Business, Columbia University, ob2105@gsb.columbia.edu

Antoine Desir

Technology and Operations Management, INSEAD, antoine.desir@insead.edu

Vineet Goyal, Garud Iyengar

Department of IE&OR, Columbia University, vg2277@columbia.edu, garud@ieor.columbia.edu

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One of the central challenges in online advertising is attribution, namely, assessing the contribution of individual advertiser actions such as e-mails, display ads and search ads to eventual conversion. Several heuristics are used for attribution in practice; however, most do not have any formal justification. The main contribution in this work is to propose an axiomatic framework for attribution in online advertising. We show that the most common heuristics can be cast under the framework and illustrate how these may fail. We propose a novel attribution metric, that we refer to as *counterfactual adjusted Shapley value* (CASV), which inherits the desirable properties of the traditional Shapley value while overcoming its shortcomings in the online advertising context. We also propose a Markovian model for the user journey through the conversion funnel, in which ad actions may have disparate impacts at different stages. We use the Markovian model to compare our metric with commonly used metrics. Furthermore, under the Markovian model, we establish that the CASV metric coincides with an adjusted “unique-uniform” attribution scheme. This scheme is efficiently implementable, and can be interpreted as a correction to the commonly used uniform attribution scheme. We supplement our theoretical developments with numerical experiments using a real-world large-scale dataset.

Key words. Digital economy, online advertising, attribution, Markov chain, Shapley value, causality

1. Introduction

With the rise of the Internet, the digital economy has become a trillion dollar industry accounting for over 6% of the U.S. GDP (Hagan 2018). Retailers reach consumers through different online channels with the goal of acquiring new customers and managing relationships with the existing ones. The wide range and frequency of economic activities taking place in the digital world has enabled data collection at a massive scale, allowing retailers to better understand customer behavior and improve service quality using data-driven decisions. With over a billion people with access to the Internet, it is not surprising that advertising has moved to the digital space. The global digital marketing sector witnessed a growth of 21% in 2017, which increased its market size to

USD 88 billion (Sluis 2018). Key decisions in this space include allocating budget to various ad channels and media (e-mail, display media platforms, and paid search for instance) and optimizing tactical decisions in each channel. Such tactical decisions may be driven by the publishers; see, e.g., Balseiro et al. (2014), Hojjat et al. (2017) and Lejeune and Turner (2019) or by the advertisers themselves. For example, in display or search advertising, this would entail bidding for ads to push one's product towards the desired customer demographic at the right time; see, e.g., Iyer et al. (2014), Balseiro et al. (2015), Balseiro and Gur (2017) and Baardman et al. (2019). Furthermore, due to the digital nature of such online ad exchanges, advertisers can access user-level information before placing a bid, which motivates the importance of understanding the *state-specific* value of an advertiser action (ad action). All such tactical decisions require a deep understanding of the value of showing an ad via a specific channel at a given time. How can an advertiser assign or *attribute* value to the advertising actions taken across the different channels and media? *Attribution* is one of the central questions in online advertising. The value of each channel is an important input to media mix optimization, helps build an understanding of the customer journey, and also helps a company justify its marketing spend (United 2012, Priest 2017). Incorrect understanding of the effectiveness of online channels can result in suboptimal budget allocation, possibly resulting in significant lost revenue (Kireyev et al. 2016). Recognizing the importance of the attribution problem, the Marketing Science Institute has consistently identified attribution as a topmost research priority over the last few years (Institute 2016, 2018, 2020). Though multiple heuristics have been proposed and studied in this domain, there is lack of a systematic approach that has both theoretical foundations and is tractable.

Attribution is inextricably linked to causality since it involves quantifying the added value of showing an ad over the baseline value of what *would have* happened if no ad was shown (*counterfactual*). We consider the following view of causality, which comes from the pioneering work of Rubin (1974):

“Intuitively, the causal effect of one treatment, E , over another, C , for a particular unit and an interval of time from t_1 to t_2 is the difference between what would have happened at time t_2 if the unit had been exposed to E initiated at t_1 and what would have happened at t_2 if the unit had been exposed to C initiated at t_1 .”

In the context of digital advertising, treatments E and C correspond to advertiser taking, and not taking an ad action, respectively. Accordingly, attribution involves capturing the causal effect of an ad where the baseline corresponds to not showing an ad.

Causality is an active research area, and there exist paradigms for capturing the causal effect of a treatment. See, e.g. Pearl (2009), Halpern and Pearl (2005), Chockler and Halpern (2004), Hitchcock (1997), Morgan and Winship (2014), Collins et al. (2004), Eells (1991), Hume (2003) and

Rubin (1974). We comment on some of these alternative approaches in our concluding remarks. In addition to these foundational or axiomatic approaches to causality, there also exist data-driven approaches, see e.g., Bottou et al. (2013), where one uses importance sampling in a Bayesian network to estimate counterfactuals, without having to collect additional data. In particular, they get a handle on the “what would have happened” scenario using the data that one has *already* collected. In Section 5.2, we leverage this technique to construct data efficient algorithms.

As with the existing work in attribution, we also consider the setting where an advertiser is interested in understanding the contributions of various ads to a single product they are promoting. Even with a single product, attribution is a challenging problem. It involves distributing the value generated by a system of actions to each individual action. The cooperative impact of the system of actions might have a mix of interaction effects that one needs to account for when decomposing the total value. Capturing such (possibly non-linear) interaction effects is a fundamental challenge.

1.1. Related literature

The attribution problem has been studied widely and in diverse areas. We present a brief overview of existing approaches and refer the reader to existing surveys (Choi et al. 2017, Kannan et al. 2016) for a more complete treatment. Attribution methodologies can be classified into two broad classes: *rule-based* or *algorithmic*. We discuss both the classifications below.

1.1.1. Rule-based heuristics. Rule-based heuristics include approaches such as *last touch attribution* (LTA), *uniform weights*, and *customized weights* (Arensman and Yeung 2016, Priest 2017, Quantcast 2013, 2016). In LTA, an advertiser attributes all the value generated by a user to the last ad action, whereas under a uniform weights scheme, all the ad actions on a conversion path are allocated an equal credit. Ad actions receive tailored weights under a custom weights scheme. Although such heuristics are transparent and tractable, there is no rationale justifying their appropriateness as a measure for attribution. Metrics such as LTA can be unfair since they do not value the contribution of channels that build product awareness. Uniform or customized weights might appear to be a fix but there is no a priori reason to believe that attribution should be linear.

1.1.2. Algorithmic approaches. The algorithmic approaches can be classified as using either *incremental value heuristic (IVH)* (or *removal effect*) or *Shapley value (SV)* as a measure for attribution.

IVH. IVH computes the change in the eventual *conversion*¹ probability of a user when a specific ad is removed from her path. This is the most common metric for attribution (Abhishek et al. 2012, Anderl et al. 2016, Arava et al. 2018, Archak et al. 2010, Danaher and van Heerde 2018,

¹ Conversion refers to the event in which a user buys the underlying product.

Kakalejčák et al. 2018, Li and Kannan 2014, Li et al. 2017). In this approach, one calibrates a model that predicts the conversion probability as a function of the ad actions and then, uses the estimated model to compute the incremental value of each action. The novelty comes from the model proposed to describe user behavior, e.g., a hidden Markov model (HMM) (Abhishek et al. 2012) or a neural network (Arava et al. 2018). There exists little (if any) formal justification for why IVH is a good attribution scheme. In Section 3.2, we show that IVH can result in unjustifiable attribution.

SV. SV (Shapley 1953) is a well-accepted concept for assigning credit to individual players in a cooperative game. The value generated by online advertising can be viewed as the outcome of a cooperative effect of the actions taken on various channels and media platforms. Dalessandro et al. (2012) pose attribution as a causal estimation problem and propose SV as an approximation scheme for the causally motivated problem. They also show that SV generalizes the probabilistic model of Shao and Li (2011). Using a stylized model, Berman (2018) shows the use of SV for attribution can be beneficial to the advertiser. Under a different stylized setting, Abhishek et al. (2017) analyze attribution contracts (including SV) used by an advertiser to incentivize two publishers that affect customer acquisition and they highlight an interesting tension between “fairness” to the publishers and “optimality” for the advertiser. Unlike in Berman (2018) and Abhishek et al. (2017), in our work, publishers are not strategic. Our work focuses on the “fairness” of the attribution to the various channels and avenues available to the advertiser. It is also worth mentioning that the online advertising industry is embracing using SV for attribution; see for example Google (2019).

Other works. Attribution has been tackled from various other angles. Jordan et al. (2011) use a Markovian model to motivate a payment scheme that satisfies incentive compatibility for the advertiser and is fair from the publisher’s point-of-view. Xu et al. (2014) propose a mutually exciting point process to capture dynamic interactions among various ads. Zhang et al. (2014), Ji et al. (2016), and Ji and Wang (2017) provide an interesting view of attribution via the lens of survival theory. Zhao et al. (2018) propose a regression-based relative importance method to compute the marginal contributions. From an empirical perspective, Blake et al. (2015) measure the return on investment of paid search and document the importance of accounting for the counterfactual.

1.2. Our approach and contributions

In spite of almost a decade of research, it remains unclear as to what is an “appropriate” or “best” attribution measure. IVH appears to be the most popular; but, to the best of our knowledge, there is no systematic framework to support it. In fact, as we later discuss, IVH suffers from serious drawbacks in certain settings. On the other hand, SV has a strong theoretical justification and a number of desirable properties such as *efficiency*, *symmetry*, *linearity*, and *null player*; in fact, SV

is the unique solution to a cooperative game that has all these properties. However, estimating SV exactly is computationally intractable in general, and one has to resort to approximations (Avrachenkov et al. 2012, Castro et al. 2009, Fatima et al. 2008, Liben-Nowell et al. 2012, Littlechild and Owen 1973, Maleki et al. 2013, Michalak et al. 2013, Owen 1972). In addition, as we show in Section 4.2, SV is not counterfactual in nature. We seek a metric that has the desirable properties of SV, and yet is tractable and able to accommodate counterfactual reasoning.

Our contributions are as follows. First, we construct an abstract Markov chain model for the customer journey through the conversion funnel that generalizes most of the existing Markovian models in the attribution literature. In every period, the customer is in one of the finitely many *states*. The advertiser observes the state and takes an *action*. The customer transitions to a random state distributed according to a probability mass function that depends on the current state and the advertising action. We propose attributing value to each state-action pair, which is a generalization of the existing approaches that attribute only to advertising actions. This extension allows us to capture state-specific attribution for each action, the need for which is supported by the finding in Bleier and Eisenbeiss (2015). We analytically quantify the additional value of state-specific attribution and show that attributing at a state-action level allows one to capture a multitude of interactions that are missed otherwise.

Second, we show that the Markov chain model allows us to generate intuitive canonical Markovian networks that serve as a robustness check for an attribution scheme. Using these networks, we show that the current attribution metrics (LTA, IVH, and uniform) have serious limitations. Furthermore, we show how one can compute state-specific SV for each action in our Markovian model and highlight that it does not adjust for the counterfactual, and hence, is not an appropriate metric for attribution in our setting. To the best of our knowledge, this is the first work in the literature to analyze the various existing attribution metrics using a common framework.

Third, and the main contribution of this work, is an axiomatic framework for attribution in online advertising that explicitly accounts for the counterfactual action. We propose a new metric for attribution that we call *counterfactual adjusted Shapley value* (CASV). We show that our proposed metric inherits the desirable axioms of the classical SV-based attribution. Note that we do not need the Markovian model to define CASV and establish the uniqueness of this attribution scheme for a given underlying (possibly non-Markovian) model. We show that this new metric leads to appropriate attribution in the Markovian networks that highlighted limitations of the existing metrics. In addition, we establish that CASV admits a crisp characterization under our Markovian model. It coincides with a *unique-uniform attribution scheme* that explicitly adjusts for the counterfactual; which, in turn, can be interpreted as a correction to the commonly used uniform attribution scheme. Furthermore, we exploit this characterization to develop simple algorithms to

estimate our metric. We demonstrate the scalability of CASV for the Markovian framework by performing numerical experiments on a real-world large-scale dataset.

Outline. The remainder of this paper is organized as follows. In Section 2, we introduce the Markovian model that describes the user journey as a function of ad exposure. In Section 3, we discuss how the existing attribution schemes (LTA, IVH, and uniform) apply to the Markovian model and we construct canonical Markovian networks that highlight the limitations of these attribution schemes. In Section 4, we introduce SV-based attribution and showcase its drawbacks. Next, we present our axiomatic framework for CASV. In Section 5, we show that the CASV-based attribution scheme is equivalent to the adjusted unique-uniform attribution scheme for the Markovian model, and propose simple algorithms to estimate it. In Section 6, we apply the proposed Markovian framework and the CASV-based attribution scheme to a large-scale real-world dataset. In Section 7, we discuss the value of state-specific attribution. We conclude in Section 8 with some directions for ongoing and future research.

2. Model

We propose a Markovian model for user behavior where transitions in user’s state are stochastic, and are a function of only the current state and the advertiser action. This process ends when the user either quits (leaves the system) or converts (buys the product). In Section 2.1, we define the components of the Markov chain (denoted by $\mathcal{M}$). When constructing the Markov chain, we keep most of its elements abstract to showcase its flexibility. In Section 2.2, we show how many practical settings can be modeled using the proposed Markovian framework. We conclude this section by defining the attribution problem for our Markovian model.

We note that our model builds upon existing works in this literature that have also used a Markovian model to describe user behavior (Abhishek et al. 2012, Anderl et al. 2016, Kakalejčik et al. 2018). We define our model in abstract terms and therefore, most of the existing models can be seen as special instances of our model.

2.1. Markovian model of user behavior

We first discuss the state space of the Markov chain, followed by the arrival process of the users, then the action space of the advertiser, and finally, we define the transition probabilities.

State space. We define $\mathbb{S} := \{s\}_{s=1}^m$ as the set of states excluding the two absorbing states (quit q and conversion c) and $\mathbb{S}^+ := \mathbb{S} \cup \{q, c\}$. The conversion state c refers to the user purchasing the underlying product, and the quit state q indicates that the user decided not to buy the product and left the system². In order to highlight the flexibility of our model, we do not give a concrete meaning to the non-absorbing states. We discuss some examples in Section 2.2.

² To tie it with the existing literature, it can be seen as the user “leaving the system” (Abhishek et al. 2017).

Arrival process. External traffic arrives at state $s \in \mathbb{S}$ w.p. λ_s (*initial state probability*). We define the vector $\boldsymbol{\lambda} \in \mathbb{R}^m$ as $[\lambda_s]_{s \in \mathbb{S}}$. We assume no external traffic arrives at c and q .

Action space. We define $\mathbb{A} := \{a\}_{a=1}^n$ as the set of advertiser actions, such as sending an e-mail or showing a display ad. We include a baseline or no-ad action ($a = 1$)³ in $\mathbb{A}$ that captures the impact of both the baseline online actions and any offline promotion efforts of the advertiser that have an impact on online transitions. Let β_s^a denote the probability that an advertiser takes action $a \in \mathbb{A}$ at state $s \in \mathbb{S}$. We denote by $\boldsymbol{\beta}$ the collection of all β_s^a values and assume it is fixed.

Transition probabilities. We denote by $p_{ss'}^a$ the probability a user moves from $s \in \mathbb{S}$ to $s' \in \mathbb{S}$ in one transition given the advertiser takes action $a \in \mathbb{A}$ at s . Also, for all $(s, s') \in \mathbb{S}^2$, we define $p_{ss'}^\beta := \sum_{a \in \mathbb{A}} \beta_s^a p_{ss'}^a$, which denotes the *average* transition probability. To keep the notation concise, for each $a \in \mathbb{A}$, we define the matrix $\mathbf{P}^a := [p_{ss'}^a]_{(s,s') \in \mathbb{S}^2} \in \mathbb{R}^{m \times m}$. Furthermore, $\mathbf{P}^\beta := [p_{ss'}^\beta]_{(s,s') \in \mathbb{S}^2} \in \mathbb{R}^{m \times m}$ and $\mathbf{B}^a := \text{diag}([\beta_s^a]_{s \in \mathbb{S}}) \in \mathbb{R}^{m \times m}$ for each $a \in \mathbb{A}$ is a diagonal matrix. Clearly, $\mathbf{P}^\beta$ represents the transition matrix over the *partial* state space $\mathbb{S}$ and $\mathbf{P}^\beta = \sum_{a \in \mathbb{A}} \mathbf{B}^a \mathbf{P}^a$. For all $s \in \mathbb{S}$, we define the vector $\mathbf{p}_s^a \in \mathbb{R}^m$ as the s -th row of $\mathbf{P}^a$ for all $a \in \mathbb{A}$ and $\mathbf{p}_s^\beta \in \mathbb{R}^m$ as the s -th row of $\mathbf{P}^\beta$. Next, we state the only assumption we make on our problem primitives.

Assumption 1 (Absorption). *Any Markov chain corresponding to $(\boldsymbol{\lambda}, \mathbf{P}^\beta)$ is absorbing, i.e., the probability each user will eventually either quit or convert from any state equals 1.*

Assumption 1 is equivalent to saying that from any state, there exists a positive probability path to one of the absorbing states. We verify this condition in the context of our numerical study (Section 6).

So far, we have not discussed the transitions to and from states q and c . We use the notation p_{sc}^a and p_{sc}^β to denote the action-specific and average one-step transition probabilities from $s \in \mathbb{S}$ to c for all $a \in \mathbb{A}$. (For transitions from $s \in \mathbb{S}$ to q , replace the index c by q .) Since both q and c are absorbing states, the transitions from them are self-loops w.p. 1. We define $\mathbf{p}_c^a := [p_{sc}^a]_{s \in \mathbb{S}} \in \mathbb{R}^m$ for each $a \in \mathbb{A}$ and $\mathbf{p}_c^\beta := [p_{sc}^\beta]_{s \in \mathbb{S}} \in \mathbb{R}^m$. (Note that $\mathbf{p}_s^a$ and $\mathbf{p}_s^\beta$ for $s \in \mathbb{S}$ correspond to the probabilities of *leaving* s whereas $\mathbf{p}_c^a$ and $\mathbf{p}_c^\beta$ correspond to the probabilities of *entering* c .) Thus, a Markov chain $\mathcal{M}$ is specified by $(\boldsymbol{\lambda}, \mathbf{P}^\beta, \mathbf{p}_c^a, \mathbf{p}_q^a)$. Next, we shift our focus to some quantities of interest for the Markov chain $\mathcal{M}$.

Expected number of visits. Let $\mathbf{F}^\beta \in \mathbb{R}^{m \times m}$ denote the *fundamental matrix* of the Markov chain $\mathcal{M}$, i.e., the (i, j) -th entry of $\mathbf{F}^\beta$ equals the expected number of visits to state j when the initial state is i . Assumption 1 ensures that $\mathbf{F}^\beta$ exists, and $\mathbf{F}^\beta = (\mathbf{I} - \mathbf{P}^\beta)^{-1}$ (Grinstead and Snell 2012). Furthermore, define $\boldsymbol{\mu}^\beta := [\mu_s^\beta]_{s \in \mathbb{S}} \in \mathbb{R}^m$ where μ_s^β denotes the expected number of visits to state s given $\boldsymbol{\beta}$. It is easy to show that $(\boldsymbol{\mu}^\beta)^\top = \boldsymbol{\lambda}^\top \mathbf{F}^\beta$.

³ We are reserving $a = 0$ for future use as it will become transparent in Section 4.2.

Eventual conversion probability. We define h_s^β as the probability of eventually being absorbed in c from state $s \in \mathbb{S}$ and the vector $\mathbf{h}^\beta := [h_s^\beta]_{s \in \mathbb{S}} \in \mathbb{R}^m$. It is easy to show that $\mathbf{h}^\beta = \mathbf{P}^\beta \mathbf{h}^\beta + \mathbf{p}_c^\beta$. Thus, $\mathbf{h}^\beta = \mathbf{F}^\beta \mathbf{p}_c^\beta$. We set $h_q^\beta = 0$ and $h_c^\beta = 1$.

2.2. Model discussion

We have defined the states and actions for the Markov chain $\mathcal{M}$ in abstract terms. Next, we provide some possible settings for these quantities. These are, by no means, the only possible mappings. Indeed, we expect the specific definitions of states and actions to be context dependent.

One possibility for the state is a summary of past interactions with the customer, e.g., the number of visits to the product website, number of e-mails received, number of e-mails opened, number of display ads seen, number of display ads clicked, etc. These counts will need to be suitably quantized to control the size of the state space. The advertiser's actions could be to do nothing, or send an e-mail, or show a display ad, or bid for a search ad.

Another possibility for the state space is $\mathbb{S} = \{\text{unaware, aware, interest, desire}\}$ associated with the conversion funnel used in the marketing literature (Strong 1925, Howard and Sheth 1969, Barry 1987, Bettman et al. 1998, Court 2009, Elzinga et al. 2009, Kotler and Armstrong 2010, Mulpuru 2011, Jansen and Schuster 2011, Bruce et al. 2012), which captures the journey of the user from being unaware of the product to becoming interested, and finally purchasing. Given the fine granularity of data now available to advertisers, we believe the advertiser can infer such states using an appropriate statistical model. The action space remains the same as above.

The conversion state c refers to the user purchasing the underlying product and is easily observable in data. The transition to the quit state q indicates that the user has decided not to buy the product. In practice, the quit state is usually not explicitly observed in the data, but can be inferred from the customer's inactivity level. In our experiments with real data detailed in Section 6, we present one simple heuristic to infer whether the customer has quit.

In order to contextualize the arrival process and the transition dynamics, it is useful to consider an advertising campaign that is run for a fixed time period. The initial state of a user is her state on the first day of the campaign. Different customers can have different initial states (depending on the level of prior engagement with the product for example) and such heterogeneity is captured by our arrival process. During the advertising campaign, the state of the customer evolves dynamically as a (stochastic) function of the ads she is exposed to, which is captured by our transition probabilities. (We provide an example based on real data when we present our numerical study in Section 6.)

We note that the definitions of state and action spaces can be customized as per the needs of the advertiser (since our attribution framework is developed for the abstract Markovian model). The state space is likely context-specific and driven by the user features that are relevant to a particular

advertiser. For example, one can even encode the time dimension in the state if one wishes to explicitly model time. Learning context-specific state aggregation and Markov chain dynamics from data is an active area of research. We refer the reader to Hallak et al. (2013) and references therein.

2.3. The attribution problem

We assume that the sale of one unit of the product generates a value of 1. As a result, the total value of the system of actions taken at various states equals $\lambda^\top \mathbf{h}^\beta$. The goal of an attribution model is to allocate this value to the underlying state-action pairs⁴ in the system while accounting for the counterfactual of not taking the particular action in the state. Note that in our setting, the state-action probability β_s^a for all $(s, a) \in \mathbb{S} \times \mathbb{A}$ is fixed, i.e., we do not consider the change in β_s^a as a function of an attribution scheme. (Unlike the setting in Berman (2018) and Abhishek et al. (2017), the publishers here are *not* strategic.) Our focus here is on developing a framework to attribute the total generated value to each state-action pair in a manner that fairly accounts for the contribution of each state-action to the eventual conversion of the user. Let π_s^a denote the expected value of the attribution that the pair $(s, a) \in \mathbb{S} \times \mathbb{A}$ receives on any sample path through the Markov chain. Therefore, we call an attribution scheme *budget-feasible* if the total attribution across all state-action pairs does not exceed the total expected value generated per sample path, i.e., $\sum_{s \in \mathbb{S}} \sum_{a \in \mathbb{A}} \pi_s^a \leq \lambda^\top \mathbf{h}^\beta$. A stronger requirement is *budget-balanced*, which enforces $\sum_{s \in \mathbb{S}} \sum_{a \in \mathbb{A}} \pi_s^a = \lambda^\top \mathbf{h}^\beta$. As we will see, not all existing heuristics are budget-feasible, let alone budget-balanced.

3. Current attribution approaches and their limitations

Next, we consider commonly used attribution schemes, e.g., LTA, IVH, and uniform, in the Markov chain setting, and illustrate their limitations. As we will see, our abstract Markov chain model is a useful tool for constructing examples that illustrate these limitations. At a high-level, both LTA and uniform are “backward looking” in the sense that they split the value of a path after observing its realization (from the initial state to the end). LTA allocates all the value to the last interacting state-action pair whereas uniform allocates it equally to all the state-action pairs that appeared in the path to conversion. In contrast, IVH is “forward looking” since it attributes to a given state-action pair based on what is expected to happen in the future. It does not require the knowledge of how the actual path will unfold. All it requires is a model that can predict the change in eventual conversion probability that results if the state-action pair is removed from the path.

In many examples to come, we will sometimes assume that the no-ad action leads to the quit state with probability 1 (this could be seen as an extreme case in which the act of not advertising leads

⁴ One might want to attribute value to actions as opposed to state-action pairs. However, we attribute to state-action pairs because the impact of an action is likely to be state dependent.

a customer to a competitor). Our theory does not require such an assumption; we do so in order to highlight specific effects in the examples with the least complexity. The networks in Examples 1 to 4 have zero counterfactual value since the no-ad action leads to the quit state with probability 1. The first concrete instance of counterfactual value is in Example 5. It is straightforward to introduce counterfactual value in Examples 1 to 4 (and show the same limitations of existing approaches hold); however, doing so introduces unnecessary complexity.

3.1. Last touch attribution (LTA)

In LTA, all the value generated due to a purchase is attributed to the last state-action pair in the path. Let the path $\mathcal{P} = \{(s_1, a_1), (s_2, a_2), \dots\}$ denote a sequence of state-action pairs before the sample path terminates at one of the absorbing states $\{q, c\}$. For $(s, a) \in \mathbb{S} \times \mathbb{A}$, define

$$w_s^{a, \text{LTA}}(\mathcal{P}) := \begin{cases} 1 & \text{if } \mathcal{P} \text{ converts and } (s, a) \text{ is the last state-action pair in } \mathcal{P} \\ 0 & \text{otherwise.} \end{cases}$$

In other words, (s, a) receives complete credit if it is the last state-action pair before the convert state c on a path $\mathcal{P}$. Therefore, LTA is clearly budget-balanced and easy to implement. Under our Markov model $\mathcal{M}$, the attribution to $(s, a) \in \mathbb{S} \times \mathbb{A}$ under the LTA scheme equals

$$\pi_s^{a, \text{LTA}} := \mathbb{E}_{\mathcal{P} \sim \mathcal{M}} [w_s^{a, \text{LTA}}(\mathcal{P})].$$

However, LTA appears “unfair” since state-action pairs are not rewarded for moving the user up in the conversion funnel. We demonstrate this limitation next.

Example 1 (LTA is unfair). Consider the network in Figure 1 with self-loop probability $p = 0$ (line). Under LTA, all the value goes to the ad action at state m , i.e., $\pi_m^{2, \text{LTA}} = 1$ and $\pi_s^{a, \text{LTA}} = 0$ for all other pairs of (s, a) , even though the ad actions at the first $m - 1$ states are essential to move the user to state m , because without them the user would have quit.

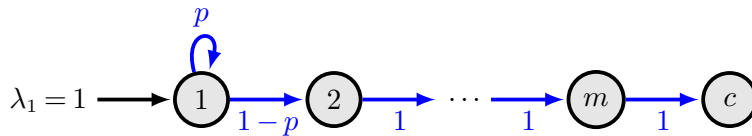


Figure 1 Network for Examples 1, 2, and 4. The action space consists of two actions: no-ad action ($a = 1$) and ad action ($a = 2$). The solid blue lines show transitions corresponding to the ad action. We assume the no-ad action in every state leads to a transition to the quit state w.p. 1. The advertiser takes the ad action at all states w.p. 1.

3.2. Incremental value heuristic (IVH)

IVH allocates to each state-action pair (s, a) the increase in the eventual conversion probability by taking action a in state s as opposed to the no-ad action. For each $a \in \mathbb{A}$, we first define an auxiliary variable that captures the corresponding forward looking increment *conditioned* on the user being at a given state:

$$\mathbf{z}^{a, \text{IVH}} := \underbrace{P^a \mathbf{h}^\beta + \mathbf{p}_c^a}_{\text{action } a} - \underbrace{(P^1 \mathbf{h}^\beta + \mathbf{p}_c^1)}_{\text{no-ad action}} = \underbrace{(P^a - P^1) \mathbf{h}^\beta}_{\text{eventual}} + \underbrace{(\mathbf{p}_c^a - \mathbf{p}_c^1)}_{\text{immediate}},$$

where $\mathbf{z}^{a, \text{IVH}} = [z_s^{a, \text{IVH}}]_{s \in \mathbb{S}} \in \mathbb{R}^m$. The scalar $z_s^{a, \text{IVH}}$ can be interpreted as the allocation at a “trace” level, i.e., if the advertiser observes a user at state s and decides to take action a , the corresponding allocation would be $z_s^{a, \text{IVH}}$. However, we need to scale this metric for it to be seen as attribution over the entire population. In particular, given that at state s , the expected number of visits is μ_s^β and action a is taken w.p. β_s^a , the IVH attribution is given by

$$\pi_s^{a, \text{IVH}} := \mu_s^\beta \beta_s^a z_s^{a, \text{IVH}}. \quad (1)$$

Although IVH is tractable and somewhat adjusts for the counterfactual, a serious limitation is that it can distribute more value than the system generates (i.e., not budget-feasible) because it pays for both eventual and immediate conversions; consequently, each conversion may be accounted for several times. We now show that this is, indeed, possible.

Example 2 (IVH can over-allocate). Consider the network in Figure 1 with $p = 0$. Under IVH, the ad action at all states receives an attribution of 1 (i.e., $\pi_s^{2, \text{IVH}} = 1$ for $s \in \{1, \dots, m\}$) and the no-ad action at all states receives an attribution of 0 (i.e., $\pi_s^{1, \text{IVH}} = 0$ for $s \in \{1, \dots, m\}$), resulting in a total allocation of m even though the value generated equals 1.

A common workaround to ensure IVH is budget-balanced is to normalize the output by an appropriate constant (*normalized IVH*). Normalizing the numbers in Example 2 results in the ad action at each state receiving $1/m$, which appears reasonable. However, as we illustrate in Example 3, even normalized IVH can be problematic.

Example 3 (Normalized IVH can be unjustified). Consider the network in Figure 2. IVH attributes 1 to the ad action at state 1 (i.e., $\pi_1^{2, \text{IVH}} = 1$) and $1/2$ to the ad action at all other states (i.e., $\pi_s^{2, \text{IVH}} = 1/2$ for $s \in \{2, \dots, m\}$). Thus, attribution under normalized IVH (denoted by $\pi_s^{a, \text{nIVH}}$) equals

$$\pi_s^{a, \text{nIVH}} = \begin{cases} \frac{1}{1+(m-1)/2} & \text{for } (s, a) = (1, 2) \\ \frac{1/2}{1+(m-1)/2} & \text{for } (s, a) \in \{(s, 2)\}_{s=2}^m \\ 0 & \text{for all other } (s, a) \text{ pairs.} \end{cases}$$

As $m \rightarrow \infty$, all the value goes to the ad action at states 2 to m , i.e., $\sum_{s=2}^m \pi_s^{2, \text{nIVH}} = 1$ and $\pi_1^{2, \text{nIVH}} = 0$ as $m \rightarrow \infty$. This appears inappropriate as the ad action at state 1 “deserves” at least half the total value since 50% of the users convert immediately after seeing the ad at state 1.

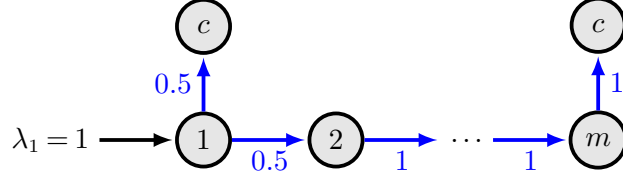


Figure 2 Network for Example 3. The action space consists of two actions: no-ad action ($a = 1$) and ad action ($a = 2$). The solid blue lines show transitions corresponding to the ad action. The no-ad action in every state leads to a transition to the quit state w.p. 1. The advertiser takes the ad action at all states w.p. 1.

3.3. Uniform attribution

Under a uniform attribution scheme, the credit generated is attributed equally to all the state-action pairs that are encountered in a path. Let the path $\mathcal{P} = \{(s_1, a_1), (s_2, a_2), \dots\}$ denote a sequence of state-action pairs before the sample path terminates at one of the absorbing states $\{q, c\}$. For $(s, a) \in \mathbb{S} \times \mathbb{A}$, define

$$w_s^{a, \text{uni}}(\mathcal{P}) := \begin{cases} \frac{n_s^a}{|\mathcal{P}|} & \text{if } \mathcal{P} \text{ converts and } (s, a) \in \mathcal{P} \\ 0 & \text{otherwise,} \end{cases}$$

where n_s^a equals the number of times (s, a) appears in $\mathcal{P}$ and $|\mathcal{P}|$ denotes the number of (not necessarily unique) state-action pairs in $\mathcal{P}$. In other words, (s, a) receives an “equal cut” of each *converted* path it contributes to. Such a scheme is budget-balanced and scalable. Under our Markov model $\mathcal{M}$, the uniform attribution for $(s, a) \in \mathbb{S} \times \mathbb{A}$ is

$$\pi_s^{a, \text{uni}} := \mathbb{E}_{\mathcal{P} \sim \mathcal{M}} [w_s^{a, \text{uni}}(\mathcal{P})]. \quad (2)$$

On the line network (Figure 1 with $p = 0$), it attributes $1/m$ to the ad action at each state (i.e., $\pi_s^{2, \text{uni}} = 1/m$ for $s \in \{1, \dots, m\}$), which seems reasonable. However, in general, it does not account for the counterfactual. Furthermore, it can result in “unfair” attribution, which we show next.

Example 4 (Uniform attribution can be unfair). Consider the network in Figure 1 with⁵ $p \in (0, 1)$. For a given path with n_1 occurrences of state-action pair $(1, 2)$, the uniform attribution scheme attributes $n_1/(n_1 + m - 1)$ to the state-action pair $(1, 2)$. As p increases, the state-action pair $(1, 2)$ receives more credit. However, this seems inappropriate since all the states 1 through m contribute equally to every conversion – just that state 1 takes several attempts to move the user towards conversion. Note that attribution measures the contribution of each state-action pair (s, a) towards conversion, and not the “effort” required.

Remark 1 (Unique-uniform). A simple fix to the above drawback is to attribute based on the number of unique state-action pairs. For instance, in the example above, the *unique-uniform* scheme

⁵ Note that a positive value of p can be interpreted as the ad action at state 1 not changing the state (e.g., awareness level) for a fraction of the user population, e.g., users who do not pay attention to promotional e-mails.

would attribute $1/m$ to the ad action at each state. However, this scheme still does not account for the counterfactual, and there is no formal rationale for it. We show later that uniform allocation adjusted for uniqueness and the counterfactual does, in fact, have a strong mathematical rationale and will be part of our prescribed method.

To summarize this section, the above examples highlight some limitations of existing heuristics but also point to the fact that no single heuristic “dominates the other.” In some way, IVH possesses a counterfactual form giving it some practical appeal. At the same time, it is only forward looking, ignoring all past actions and their potential contributions. Uniform, on the other hand, accounts for past actions, but is not counterfactual. Ideally, an attribution scheme would provide the best of both worlds, properly accounting for past actions while also accounting for the counterfactual associated with not advertising. We also note that our proposed Markov chain model can be used to generate simple canonical examples to test the robustness and intuitive appeal of attribution schemes.

4. Shapley value (SV)

In this section, we investigate SV-based attribution schemes using our Markov model. SV is a particular attractive metric for attribution in our context, since the value, i.e., customer conversion, is the result of the cooperative impact of actions taken during the course of the customer’s journey. We first present a primer on SV (Section 4.1) followed by a discussion on why a direct application to our context does not suffice (Section 4.2). Next, we propose a counterfactual adjusted SV (CASV) metric for attribution (Section 4.3) and evaluate the performance of CASV on all the motivating examples (Section 4.4). In this section, we hope to convince the reader that, leaving computational tractability aside, CASV is an attractive metric for attribution.

4.1. SV primer

SV is solution concept in coalitional game theory (Shapley 1953). Consider a finite set $\mathbb{P}$ of players with certain fixed strategies. Let the *characteristic function* $v(\mathcal{X})$ denote the value generated by coalition $\mathcal{X} \subseteq \mathbb{P}$. The value $v(\emptyset)$ of the empty coalition is normalized to 0. The goal is to “fairly” distribute the value $v(\mathbb{P})$ to the individual players. Note that the players are not assumed to be strategic here, i.e., they do not choose strategies as a function of the distribution they are likely to receive; the strategies are fixed, ex-ante.

SV distributes the value $v(\mathbb{P})$ of the grand coalition to a player $r \in \mathbb{P}$ as follows:

$$\pi_r^{\text{Shap}} := \sum_{\mathcal{X} \subseteq \mathbb{P} \setminus \{r\}} w_{|\mathcal{X}|} \times \{v(\mathcal{X} \cup \{r\}) - v(\mathcal{X})\},$$

where

$$w_{|\mathcal{X}|} := \frac{|\mathcal{X}|!(|\mathbb{P}| - |\mathcal{X}| - 1)!}{|\mathbb{P}|!}.$$

The attractiveness of SV is rooted in the fact that it is the *unique* solution concept that satisfies the following four desirable properties:

1. **Efficiency:** $\sum_{r \in \mathbb{P}} \pi_r = v(\mathbb{P})$.
2. **Symmetry:** Consider players $r, r' \in \mathbb{P}$ such that for any $\mathcal{X} \subseteq \mathbb{P} \setminus \{r, r'\}$, r and r' are *equivalent*, i.e., $v(\mathcal{X} \cup \{r\}) = v(\mathcal{X} \cup \{r'\})$. Then, $\pi_r = \pi_{r'}$.
3. **Linearity:** Consider two characteristic functions $v_1(\cdot)$ and $v_2(\cdot)$. Linearity states that for all players $r \in \mathbb{P}$, $\pi_r(v_1 + v_2) = \pi_r(v_1) + \pi_r(v_2)$ and $\pi_r(\alpha v_1) = \alpha \pi_r(v_1)$ for all $\alpha \in \mathbb{R}$.
4. **Null player:** Suppose player $r \in \mathbb{P}$ does not add any value to any coalition, i.e., for all $\mathcal{X} \subseteq \mathbb{P} \setminus \{r\}$, $v(\mathcal{X} \cup \{r\}) = v(\mathcal{X})$. Then, $\pi_r = 0$.

4.2. Direct application

SV is a natural candidate for decomposing the system value because the state-action pairs can be viewed as players participating in a cooperative game to achieve a common goal of converting the users. In particular, given the action intensities β , the Markovian system generates a value of $\lambda^\top h^\beta$ and the state-action-specific SV $\pi_s^{a, \text{Shap}}$ represents the credit allocated to each state-action pair $(s, a) \in \mathbb{S} \times \mathbb{A}$. We first define the underlying components (players, coalitions, and the characteristic function) in our context and then discuss an important drawback of such a naïve application.

Player. We want to allow for the possibility that the value of an action $a \in \mathbb{A}$ is state-dependent, and thus want to attribute to state-action pairs (s, a) . In order to make this possible in the SV setting, we must define the set of players $\mathbb{P} = \mathbb{S} \times \mathbb{A}$. Note that we want to allow attribution to the baseline no-ad action ($a = 1$), i.e., we are explicitly interested in understanding the value of the baseline action.

Coalition. Next, we need to define the value associated with a coalition $\mathcal{X} \subseteq \mathbb{P} \equiv \mathbb{S} \times \mathbb{A}$. In the usual application of SV, one computes the value $v(\mathcal{X})$ generated by the coalition $\mathcal{X}$, assuming that the rest of the players $\mathbb{P} \setminus \mathcal{X}$ are *absent*, i.e., not contributing to the value generation process in any manner. In our setting, we accomplish this by postulating that the customer exits the system when an action a is employed in state s and the pair $(s, a) \notin \mathcal{X}$. We operationalize this by defining a “zero-value action” ($a = 0$) that transitions customers to the quit state with probability 1, and setting the frequency $\beta_s^0 = \sum_{a: (s, a) \notin \mathcal{X}} \beta_s^a$, i.e., we take action 0 in state s whenever we intended to take action a , but $(s, a) \notin \mathcal{X}$. Note that the zero-value action is merely a construct to define the system behavior at $(s, a) \notin \mathcal{X}$, and is never an element of any coalition, and therefore, does not receive any attribution. Let $\beta^\mathcal{X}$ denote the state-action frequencies corresponding to the coalition

$\mathcal{X}$ where the effective set of actions is $\mathbb{A} \cup \{0\}$. Note that when $\mathcal{X} = \emptyset$, the zero-value action is taken with probability 1 in all states, and therefore, $v(\emptyset) = 0$.

We note here that an alternative approach would be to attribute only to actions $a \neq 1$, i.e., actions other than the baseline no-ad action. In this setting, the set of players $\mathbb{P} = \mathbb{S} \times (\mathbb{A} \setminus \{1\})$, and we take the baseline action 1 in state s whenever we intended to take action a , but $(s, a) \notin \mathcal{X}$. In this setting, the value of the empty coalition corresponds to the value generated by taking only baseline no-ad action in all states, and this value may not be zero in our setting because of offline promotional activities, i.e., in this specification, $v(\emptyset) \neq 0$ is possible. Since SV allocates the additional value $v(\mathcal{X}) - v(\emptyset)$ to $\mathcal{X} \subseteq \mathbb{P} = \mathbb{S} \times (\mathbb{A} \setminus \{1\})$, the actions $a \neq 1$ are allowed to “free ride” on the value generated by the baseline action. Our approach of introducing the zero-value action, and explicitly attributing value to the baseline no-ad action prevents this free riding. See Appendix A for a more detailed discussion.

Characteristic function. Given $\mathcal{X} \subseteq \mathbb{S} \times \mathbb{A}$, we define $v(\mathcal{X})$ to be the conversion probability when the state-action frequencies are given by $\beta^{\mathcal{X}}$. For the Markov chain model, $v(\mathcal{X}) := \lambda^\top h^{\beta^{\mathcal{X}}}$. Since the value depends on the model $\mathcal{M}$, we will use the notation $v_{\mathcal{M}}(\cdot)$ when the emphasis on $\mathcal{M}$ is necessary.

Shapley value. The SV for each $(s, a) \in \mathbb{S} \times \mathbb{A}$ is

$$\pi_s^{a, \text{Shap}}(v) := \sum_{\mathcal{X} \subseteq \{\mathbb{S} \times \mathbb{A}\} \setminus \{(s, a)\}} w_{|\mathcal{X}|} \times \{v(\mathcal{X} \cup \{(s, a)\}) - v(\mathcal{X})\}, \quad (3)$$

where

$$w_{|\mathcal{X}|} := \frac{|\mathcal{X}|!(mn - |\mathcal{X}| - 1)!}{(mn)!}. \quad (4)$$

SV depends on our choice of the characteristic function $v(\cdot)$ and the model $\mathcal{M}$ and hence, we will use the notation $\pi_s^{a, \text{Shap}}(v)$ and $\pi_s^{a, \text{Shap}}(\mathcal{M})$ when the emphasis on $v(\cdot)$ and $\mathcal{M}$ is necessary. Though this view of attribution inherits the desirability of SV, it suffers from a critical flaw: it does not adjust for the counterfactual (Example 5).

Example 5 (Need for counterfactual). Consider the network in Figure 3. Showing an ad in state 1 should *not* get any attribution since it provides no *additional* value over the counterfactual action (no ad). However, (3) attributes all the value to the ad action as shown below:

$$\begin{aligned} \pi_1^{2, \text{Shap}} &= \underbrace{\frac{1}{2} \{v(\{(1, 2)\}) - v(\emptyset)\}}_{\text{“}\mathcal{X}=\emptyset\text{”}} + \underbrace{\frac{1}{2} \{v(\{(1, 2), (1, 1)\}) - v(\{(1, 1)\})\}}_{\text{“}\mathcal{X}=\{(1, 1)\}\text{”}} = \frac{1}{2} \{1 - 0\} + \frac{1}{2} \{1 - 0\} = 1 \\ \pi_1^{1, \text{Shap}} &= \underbrace{\frac{1}{2} \{v(\{(1, 1)\}) - v(\emptyset)\}}_{\text{“}\mathcal{X}=\emptyset\text{”}} + \underbrace{\frac{1}{2} \{v(\{(1, 1), (1, 2)\}) - v(\{(1, 2)\})\}}_{\text{“}\mathcal{X}=\{(1, 2)\}\text{”}} = \frac{1}{2} \{0 - 0\} + \frac{1}{2} \{1 - 1\} = 0. \end{aligned}$$

Note that $v(\{(1, 1)\}) = 0$ since the action intensity of the no-ad action β_1^1 is 0 in this example. Furthermore, even LTA and uniform attribution fail this sanity check.

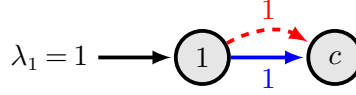


Figure 3 Network for Example 5. The action space consists of two actions: no-ad action ($a = 1$) and ad action ($a = 2$). Solid blue (dashed red) lines denotes transitions for the ad action (no-ad action). The advertiser takes the ad action at state 1 w.p. 1, i.e., $\beta_1^2 = 1$ and $\beta_1^1 = 0$.

4.3. Counterfactual adjusted Shapley value (CASV)

As we alluded to earlier, we seek a measure that provides the best of two worlds: (1) appropriately captures the contributions of the past actions (a property of, e.g., uniform and SV)⁶ and (2) exhibits counterfactual reasoning (capturing a feature of IVH). To this end, we focus on adapting SV to account for the counterfactual with respect to the baseline no-ad action and we do so by adhering to the causality framework of Rubin (1974). We first define a *counterfactual player* and then define the *counterfactual adjusted Shapley value* (CASV). Next, we motivate the desirability of CASV.

Counterfactual player. Following Rubin (1974), at state s , the counterfactual to taking action a w.p. β_s^a is to take the no-ad action (action 1) w.p. β_s^a (in addition to the default intensity of β_s^1). Accordingly, for a given player $(s, a) \in \mathbb{S} \times \mathbb{A}$, we denote the counterfactual player as $(s, 1)^a$, where the “ a ” in the superscript captures the dependence on β_s^a .

Counterfactual adjusted Shapley value. The game theoretic setup (player, coalition, characteristic function) is the same as in Section 4.2. For each $(s, a) \in \mathbb{S} \times \mathbb{A}$, we define CASV as

$$\psi_s^{a, \text{Shap}} := \sum_{\mathcal{X} \subseteq \{\mathbb{S} \times \mathbb{A}\} \setminus \{(s, a)\}} w_{|\mathcal{X}|} \times \{v(\mathcal{X} \cup \{(s, a)\}) - v(\mathcal{X} \cup \{(s, 1)^a\})\}, \quad (5)$$

where $w_{|\mathcal{X}|}$ is the same as in (4). We use the symbol ψ instead of π to differentiate CASV from SV. The only change we make in going from $\pi_s^{a, \text{Shap}}$ to $\psi_s^{a, \text{Shap}}$ is that we replace $v(\mathcal{X})$ by $v(\mathcal{X} \cup \{(s, 1)^a\})$, i.e., CASV captures the additional value added to the coalition $\mathcal{X}$ by a player (s, a) as compared to the value added by its counterfactual player $(s, 1)^a$. Note that the definition of $v(\mathcal{X})$ assumes that state-action pairs $(s, a) \notin \mathcal{X}$ are replaced by the state-zero-action $(s, 0)$.

It is unclear whether CASV under the current cooperative game is equivalent to SV under a different cooperative game. However, CASV can be expressed as the difference between two SVs:

$$\psi_s^{a, \text{Shap}}(\mathcal{M}) = \pi_s^{a, \text{Shap}}(\mathcal{M}) - \pi_s^{a, \text{Shap}}(\mathcal{M}_{(s, a)}), \quad (6)$$

where $\mathcal{M}$ denotes the original network and $\mathcal{M}_{(s, a)}$ denotes a counterfactual network for (s, a) where we replace the state-action pair (s, a) by $(s, 1)$, i.e., take action 1 with probability $\beta_s^1 + \beta_s^a$. Here, $\pi_s^{a, \text{Shap}}(\mathcal{M}_{(s, a)})$ is the *counterfactual value* of (s, a) , i.e., the value generated if no-ad action 1 was taken instead of the ad action a at state s .

⁶ We will see in Section 5 that uniform and SV are closely linked.

Next, we present an axiomatic justification for CASV. However, one needs to be careful when defining the axioms in the counterfactual context. To be specific, efficiency should now pertain to the redistribution of additional value generated over the counterfactual value. Similarly, the definition of equivalent players should be adjusted, and the null player should correspond to a player with zero value-add. We give an axiomatic definition for CASV in Theorem 1 (see Appendix B for the proof). The relevance of these axioms in the context of online advertising is discussed in Section 4.4, where we revisit the motivating examples.

Theorem 1 (Axioms). *Fix a model $\mathcal{M}$. Then, CASV is the unique solution of the following four counterfactual axioms.*

1. **Counterfactual efficiency:** *The sum of CASVs equals the additional value generated over the counterfactual value, i.e.,*

$$\sum_{(s,a) \in \mathbb{S} \times \mathbb{A}} \psi_s^{a,Shap} = v(\mathbb{S} \times \mathbb{A}) - \sum_{(s,a) \in \mathbb{S} \times \mathbb{A}} \pi_s^{a,Shap}(\mathcal{M}_{(s,a)}).$$

2. **Counterfactual symmetry:** *If $(s,a) \in \mathbb{S} \times \mathbb{A}$ and $(s',a') \in \mathbb{S} \times \mathbb{A}$ are counterfactual equivalent, i.e.,*

$$(A) \quad v(\mathcal{X} \cup \{(s,a)\}) - v(\mathcal{X} \cup \{(s,1)^a\}) = v(\mathcal{X} \cup \{(s',a')\}) - v(\mathcal{X} \cup \{(s',1)^{a'}\}) \text{ and}$$

$$(B) \quad v(\mathcal{X} \cup \{(s,1)^a, (s',a')\}) = v(\mathcal{X} \cup \{(s',1)^{a'}, (s,a)\})$$

for all $\mathcal{X} \subseteq \{\mathbb{S} \times \mathbb{A}\} \setminus \{(s,a), (s',a')\}$, then

$$\psi_s^{a,Shap} = \psi_{s'}^{a',Shap}.$$

3. **Linearity:** *Consider two characteristic functions $v_1(\cdot)$ and $v_2(\cdot)$. For all $(s,a) \in \mathbb{S} \times \mathbb{A}$, we have*

$$\psi_s^{a,Shap}(v_1 + v_2) = \psi_s^{a,Shap}(v_1) + \psi_s^{a,Shap}(v_2)$$

and for all $\alpha \in \mathbb{R}$,

$$\psi_s^{a,Shap}(\alpha v_1) = \alpha \psi_s^{a,Shap}(v_1).$$

4. **Counterfactual null player:** *Consider a player $(s,a) \in \mathbb{S} \times \mathbb{A}$ that has a zero value-add to all coalitions that do not contain (s,a) , i.e., for all $\mathcal{X} \subseteq \{\mathbb{S} \times \mathbb{A}\} \setminus \{(s,a)\}$,*

$$v(\mathcal{X} \cup \{(s,a)\}) = v(\mathcal{X} \cup \{(s,1)^a\}).$$

Then, $\psi_s^{a,Shap} = 0$.

In Theorem 1, we define $(s,a) \in \mathbb{S} \times \mathbb{A}$ and $(s',a') \in \mathbb{S} \times \mathbb{A}$ to be counterfactual equivalent if they satisfy (A) and (B). To better understand these conditions, consider the following three conditions for all $\mathcal{X} \subseteq \{\mathbb{S} \times \mathbb{A}\} \setminus \{(s,a), (s',a')\}$:

- (A1) $v(\mathcal{X} \cup \{(s, a)\}) = v(\mathcal{X} \cup \{(s', a')\})$,
 (A2) $v(\mathcal{X} \cup \{(s, 1)^a\}) = v(\mathcal{X} \cup \{(s', 1)^{a'}\})$, and
 (B) $v(\mathcal{X} \cup \{(s, 1)^a, (s', a')\}) = v(\mathcal{X} \cup \{(s', 1)^{a'}, (s, a)\})$.

Condition (A1) states that adding either of the two players to any coalition $\mathcal{X}$, containing neither player, has the same effect on the system value, and condition (A2) states that the same is true for the corresponding counterfactual players. The only remaining case of interest is that we add the first player and the counterfactual of the second player, or the other way around. Condition (B) states that doing either is equivalent in terms of the characteristic function. Note that we only need conditions (A) and (B) for two players to be counterfactual equivalent, and (A1) and (A2) are sufficient conditions for (A). It is worth mentioning that our definition of counterfactual equivalent players is a generalization of equivalent players from Section 4.1. In particular, equivalent players only need to satisfy condition (A1) and since the notion of a counterfactual player does not exist in the SV context, conditions (A2) and (B) are irrelevant.

Remark 2 (Extension to a non-Markovian model). Theorem 1 (and hence, the uniqueness of CASV w.r.t. the counterfactual axioms) holds for any model that describes the conversion probability as a function of the state-action frequencies β . In particular, the model need not be Markovian. Our proof of Theorem 1 does not require the Markovian assumption. However, the Markovian customer model simplifies the computation of CASV, which is intractable to compute in general. We elaborate further in Section 5.

Note that the uniqueness result in Theorem 1 is for a given underlying model $\mathcal{M}$, in the sense that the attribution *can* change if the model $\mathcal{M}$ changes. In general, the attribution task involves two fundamental and challenging tasks: (1) defining an appropriate model $\mathcal{M}$ and (2) given model $\mathcal{M}$, computing the attribution. Although, we briefly discuss model selection in Section 2.2 (and provide an example in Section 6), the primary focus in this work is on the second challenge. An important complementary direction pertains to the development of model-robust attribution schemes. Ideally, one would want the attribution scheme to be robust to model perturbations, i.e., given two “close” models $\mathcal{M}_1$ and $\mathcal{M}_2$, one would want the corresponding attributions to be “close.” We leave the study of such robustness of CASV (and other existing heuristics)⁷ and the development of “model-robust” attribution schemes for future work.

We conclude this subsection with a discussion regarding the attribution to the baseline no-ad action. By definition of CASV, the baseline no-ad action at each state receives zero counterfactual credit. However, one can attribute the counterfactual value $\sum_{(s,a) \in \mathbb{S} \times \mathbb{A}} \pi_s^{a, \text{Shap}}(\mathcal{M}_{(s,a)})$ (“residual”)

⁷ We thank an anonymous referee for highlighting the challenge of attribution robustness to model specification.

to the no-ad action in a post hoc fashion. In other words, the post hoc attribution to the no-ad action at state $s \in \mathbb{S}$ equals

$$\sum_{a \in \mathbb{A}} \pi_s^{a, \text{Shap}}(\mathcal{M}_{(s,a)}) = \pi_s^{1, \text{Shap}}(\mathcal{M}) + \sum_{a=2}^n \pi_s^{a, \text{Shap}}(\mathcal{M}_{(s,a)}), \quad (7)$$

where we use the fact that $\mathcal{M}_{(s,1)} = \mathcal{M}$. The two terms in (7) have an intuitive interpretation. The first term captures the value generated by the no-ad action due to a positive value of β_s^1 whereas the second term captures the value that *would have* been generated if the no-ad action was used in place of the other actions. It is also worth highlighting that this post hoc attribution to no-ad action also captures the value that could be attributed to any advertising action not being modeled (offline ads for example). We discuss this further when we revisit Example 5 in Section 4.4.

4.4. Revisiting canonical examples

We now revisit the networks discussed so far and show that our CASV measure does not suffer from the same limitations as the other existing heuristics. We also use these relatively simple networks to illustrate the relevance of the counterfactual axioms (Theorem 1) in the online advertising context.

Examples 1, 2, and 4. For the network in Figure 1 with $p < 1$, all state-action pairs are counterfactual equivalent. Furthermore, since the counterfactual action, i.e., no-ad action, directs the user to the quit state with probability 1, there is no counterfactual value in the system. Therefore, it immediately follows that the CASV at every state equals $1/m$ for the ad action and 0 otherwise.

Example 3. For the network in Figure 2, CASV attributes 0 to the no-ad action at all the states and the following to the ad action:

$$\begin{cases} \frac{1}{2} + \frac{1}{2m} & \text{if } s = 1 \\ \frac{1}{2m} & \text{if } s \in \{2, \dots, m\}. \end{cases}$$

This seems sensible as half of the paths convert just due to the ad action at state 1 whereas the other half of the paths convert with the help of the ad action at all the m states. This example illustrates the linearity axiom. The network in Figure 2 is the “sum” of two different Markov chains $\mathcal{M}_i$, $i = 1, 2$. In chain $\mathcal{M}_1$, users convert directly after the ad at state 1, and in chain $\mathcal{M}_2$, users convert after the ad at state m , and a random user is equally likely to belong to either of the groups. The CASV on $\mathcal{M}_1$ trivially attributes a value of 1 to the ad action at state 1 and 0 to all other players, whereas the CASV on $\mathcal{M}_2$ attributes a value of $1/m$ to the ad action at each state (since it can be seen as a “line” network). The CASV for the combined network follows from linearity.

Example 5. For the network in Figure 3, CASV allocates zero credit to the ad action, which is appropriate. However, CASV of the no-ad action is, by definition, zero. So, who gets credit for the value in the system? The answer, as discussed at the end of Section 4.3, is the baseline no-ad action. Recall that the attribution to the baseline no-ad action includes attribution to all “hidden” offline actions. This network exemplifies the counterfactual null player axiom since taking an ad action does not generate any additional value over the counterfactual action of taking the no-ad action.

In sum, CASV appears to be an appealing measure for attribution: it has a number of desirable properties and appears robust to various network structures. That said, similar to SV, computing CASV using (5) requires an exponential runtime in the number of underlying players. This is an important tractability concern that could render CASV impractical in settings with moderately-sized state and action spaces. In the next section, we show that CASV can be computed efficiently for the Markovian model.

5. CASV for the Markov chain model

In this section, we characterize CASV for the Markov chain model (Section 5.1) and use the characterization to develop simple algorithms for estimating CASV (Section 5.2).

5.1. Characterization

To characterize CASV for the Markov chain model, we use the fact that CASV can be expressed as a difference of two SVs (see (6)) and hence, analyze SV first. Proposition 1 connects the coalition-oriented game-theoretic construct of SV to the paths sampled from the Markov model $\mathcal{M}$. In particular, we show that, under our Markovian setup, SV is, in fact, identical to the unique-uniform attribution scheme (motivated in Remark 1).

Proposition 1 (SV equals unique-uniform). *Consider $(s, a) \in \mathbb{S} \times \mathbb{A}$ and the Markov chain $\mathcal{M}$. The SV of (s, a) as defined in (3) equals*

$$\pi_s^{a, \text{Shap}} = \mathbb{E}_{\mathcal{P} \sim \mathcal{M}} [w_s^a(\mathcal{P})],$$

where

$$w_s^a(\mathcal{P}) := \begin{cases} \frac{1}{u(\mathcal{P})} & \text{if } \mathcal{P} \text{ converts and } (s, a) \in \mathcal{P} \\ 0 & \text{otherwise.} \end{cases}$$

The function $u(\cdot)$ returns the number of unique players and $\mathcal{P}$ is a path over players, i.e., state-action pairs.

Proof. For convenience, we define $r := (s, a)$ and $\mathbb{P} := \mathbb{S} \times \mathbb{A}$ with the understanding that $\pi_r^{\text{Shap}} := \pi_s^{a, \text{Shap}}$, and $w_r(\cdot) := w_s^a(\cdot)$. The proof is split into three parts.

Step 1: Express $v(\mathcal{X})$ as an expectation over paths:

$$\begin{aligned} v(\mathcal{X}) &= \boldsymbol{\lambda}^\top \mathbf{h}^{\beta^{\mathcal{X}}} \\ &= \boldsymbol{\lambda}^\top \left(I + P^{\beta^{\mathcal{X}}} + (P^{\beta^{\mathcal{X}}})^2 + \dots \right) \mathbf{p}_c^{\beta^{\mathcal{X}}} \\ &= \mathbb{E}_{\mathcal{P} \sim \mathcal{M}(\mathcal{X})} [\mathbb{I}_{\{\mathcal{P} \text{ converts}\}}] \\ &= \mathbb{E}_{\mathcal{P} \sim \mathcal{M}} [\mathbb{I}_{\{\mathcal{P} \text{ converts}\}} \mathbb{I}_{\{\mathcal{X}(\mathcal{P}) \subseteq \mathcal{X}\}}], \end{aligned}$$

where $\mathcal{M}(\mathcal{X})$ denotes the Markov chain in which only the players in coalition $\mathcal{X}$ are “active” and $\mathcal{X}(\mathcal{P})$ denotes the set of unique players in path $\mathcal{P}$.

Step 2: Use the representation from Step 1 in the definition of SV π_r^{Shap} as stated in (3):

$$\begin{aligned} \pi_r^{\text{Shap}} &= \sum_{\mathcal{X} \subseteq \mathbb{P} \setminus \{r\}} w_{|\mathcal{X}|} \times \{v(\mathcal{X} \cup \{r\}) - v(\mathcal{X})\} \\ &= \sum_{\mathcal{X} \subseteq \mathbb{P} \setminus \{r\}} w_{|\mathcal{X}|} \left\{ \mathbb{E}_{\mathcal{P} \sim \mathcal{M}} [\mathbb{I}_{\{\mathcal{P} \text{ converts}\}} (\mathbb{I}_{\{\mathcal{X}(\mathcal{P}) \subseteq \mathcal{X} \cup \{r\}\}} - \mathbb{I}_{\{\mathcal{X}(\mathcal{P}) \subseteq \mathcal{X}\}})] \right\} \\ &= \mathbb{E}_{\mathcal{P} \sim \mathcal{M}} [\mathbb{I}_{\{\mathcal{P} \text{ converts}\}} \times f_r], \end{aligned}$$

where

$$f_r := \sum_{\mathcal{X} \subseteq \mathbb{P} \setminus \{r\}} w_{|\mathcal{X}|} (\mathbb{I}_{\{\mathcal{X}(\mathcal{P}) \subseteq \mathcal{X} \cup \{r\}\}} - \mathbb{I}_{\{\mathcal{X}(\mathcal{P}) \subseteq \mathcal{X}\}}).$$

Step 3: Observe that for a given path $\mathcal{P}$, f_r can be seen as the SV of player r corresponding to a different game with characteristic function defined as

$$v_{\mathcal{X}(\mathcal{P})}(\mathcal{X}) := \mathbb{I}_{\{\mathcal{X}(\mathcal{P}) \subseteq \mathcal{X}\}} \text{ for all } \mathcal{X} \subseteq \mathbb{P}.$$

A game with such a characteristic function is referred to as a *carrier game* and it is well-known (see Lemma 2 of Shapley (1953) for instance) that the SV of a player in a carrier game equals $1/u(\mathcal{P})$ if the player is in $\mathcal{P}$ and 0 otherwise. Therefore,

$$f_r = \begin{cases} \frac{1}{u(\mathcal{P})} & \text{if } r \in \mathcal{P} \\ 0 & \text{otherwise,} \end{cases}$$

which implies

$$\pi_r^{\text{Shap}} = \mathbb{E}_{\mathcal{P} \sim \mathcal{M}} [w_r(\mathcal{P})].$$

This completes the proof. □

The above result enables one to characterize CASV, which constitutes the main result of this work.

Theorem 2 (CASV equals unique-uniform minus counterfactual). Consider $(s, a) \in \mathbb{S} \times \mathbb{A}$, the Markov chain $\mathcal{M}$, and the counterfactual Markov chain $\mathcal{M}_{(s,a)}$. The CASV of (s, a) as defined in (5) equals

$$\psi_s^{a, \text{Shap}} = \mathbb{E}_{\mathcal{P} \sim \mathcal{M}} [w_s^a(\mathcal{P})] - \mathbb{E}_{\mathcal{P} \sim \mathcal{M}_{(s,a)}} [w_s^a(\mathcal{P})],$$

where $w_s^a(\cdot)$ is as defined in Proposition 1.

There is a striking resemblance between this characterization of CASV and uniform attribution as defined in (2). In particular, the two expressions are identical except the definition of the weight function and the counterfactual adjustment. The weight function in CASV only rewards based on whether a state-action pair appears in the path or not (as opposed to the number of times it appears). In other words, if the ad action a was taken multiple times (say n_s^a) at the *same* state s before the user moved to another state s' , the CASV weight function only rewards it once (as opposed to rewarding it n_s^a times). This seems very reasonable and, in fact, is the simple fix we proposed in Remark 1.

It is noteworthy that the coalition-oriented construct of CASV, which on the surface does not seem to be related to the paths of the underlying Markovian model, reduces to being expressed as a remarkably simple function of such paths. This connection is quite valuable as it helps to gain deeper insights regarding the structure of CASV and hence, build a better understanding. Next, we use this connection to develop simple algorithms to estimate CASV.

5.2. Algorithms

Given a set $\mathcal{D}$ of user paths where each user path consists of various state-action-state (s, a, s') tuples, the action-specific transition probabilities and initial state probabilities can be estimated using empirical frequencies⁸. To estimate CASV for player (s, a) , one can sample paths from the estimated Markov chains $\mathcal{M}$ and $\mathcal{M}_{(s,a)}$ and use Theorem 2 directly. However, this involves sampling from a different Markov chain $\mathcal{M}_{(s,a)}$ to estimate CASV for each player, which we address next.

A simple change of measure allows CASV to be expressed as

$$\psi_s^{a, \text{Shap}} = \mathbb{E}_{\mathcal{P} \sim \mathcal{M}} \left[w_s^a(\mathcal{P}) \left(1 - \frac{g_{(s,a)}(\mathcal{P})}{g(\mathcal{P})} \right) \right], \quad (8)$$

where $g(\mathcal{P})$ and $g_{(s,a)}(\mathcal{P})$ denote the probabilities of observing path $\mathcal{P}$ under Markov chains $\mathcal{M}$ and $\mathcal{M}_{(s,a)}$, respectively. The ratio $g_{(s,a)}(\mathcal{P})/g(\mathcal{P})$ denotes the *importance weight* and it is easy to show that

$$\frac{g_{(s,a)}(\mathcal{P})}{g(\mathcal{P})} = \prod_{s' \in \mathbb{S}^+} \left(\frac{p_{ss'}^1}{p_{ss'}^a} \right)^{n_{ss'}^a(\mathcal{P})},$$

⁸ We are assuming the advertiser knows β . If not, then it can be estimated similarly.

where $n_{ss'}^a(\mathcal{P})$ denotes the number of occurrences of the tuple (s, a, s') in $\mathcal{P}$. Given (8), it suffices to sample paths from just one Markov chain ($\mathcal{M}$) to estimate CASV for all the players. In fact, each sample can be recycled for multiple players and the scheme can be implemented using a parallel (over both players and samples) architecture. We state this estimation procedure in Algorithm 1, where we assume we have access to the Markov chain $\mathcal{M}$.

In practice, the true model $\mathcal{M}$ is not known, and one is forced to use a finite sample of data to compute an estimate $\widehat{\mathcal{M}}$, and attribute according to the CASV $\psi_s^{a, \text{Shap}}(\widehat{\mathcal{M}})$ of the estimate model $\widehat{\mathcal{M}}$. We discuss sensitivity to model estimation when we report the result of our numerical experiments (Section 6). One can leverage the work of Bottou et al. (2013) to obtain confidence intervals of the estimate of CASV obtained using the importance weights technique in (8). Such confidence intervals are complementary to our sensitivity discussion and we refer the reader to Bottou et al. (2013) for more details.

Note that even if one has access to the true model $\mathcal{M}$, the output of Algorithm 1 is an *estimate* of the true CASV since it relies on Monte Carlo sampling. It is easy to show that the resulting estimate is unbiased, asymptotically consistent, and has a variance that decays at a rate of $\mathcal{O}(1/N)$, where N denotes the number of samples used in Monte Carlo. Furthermore, as we discuss in Section 6.3, we found $N = 100,000$ to be large enough to obtain a stable estimate of CASV. Accordingly, moving forward, instead of focusing on the Monte Carlo error, we will focus on the sensitivity to model estimation.

Algorithm 1 Estimating CASV given Markov chain $\mathcal{M}$

Require: Markov chain $\mathcal{M}$ and number of user paths N

```

1: Initialize:  $\psi_s^{a, \text{alg}} = 0$  for all  $(s, a) \in \mathbb{S} \times \mathbb{A}$ 
2: for  $i = 1$  to  $N$  do
3:   Sample path:  $\mathcal{P}_i \sim \mathcal{M}$ 
4:   if  $\mathcal{P}_i$  converted then
5:     for  $(s, a) \in \mathcal{P}_i$  do
6:        $\psi_s^{a, \text{alg}} = \psi_s^{a, \text{alg}} + \frac{1}{u(\mathcal{P}_i)} \left( 1 - \prod_{s' \in \mathbb{S}^+} (p_{ss'}^1 / p_{ss'}^a)^{n_{ss'}^a(\mathcal{P}_i)} \right)$ 
7:     end for
8:   end if
9: end for
10: Normalize:  $\psi_s^{a, \text{alg}} = \psi_s^{a, \text{alg}} / N$  for all  $(s, a) \in \mathbb{S} \times \mathbb{A}$ 
11: return  $\psi^{\text{alg}} := \{\psi_s^{a, \text{alg}}\}_{(s, a) \in \mathbb{S} \times \mathbb{A}}$ 

```

It is also possible to employ a Bayesian estimation approach where one maintains a belief over the transition probability vector $[p_{ss'}^a]_{s' \in \mathbb{S}^+}$ in the form of a Dirichlet distribution for each $(s, a) \in \mathbb{S} \times \mathbb{A}$.

(One can also maintain a similar belief over the initial state probabilities vector λ and action intensities vector $[\beta_s^a]_{a \in \mathbb{A}}$ for all $s \in \mathbb{S}$ if required.) Since Dirichlet and multinomial are conjugate distributions, the posterior belief can be computed easily. The posterior belief can be used to generate posterior samples for CASV as follows. Use the posterior belief to generate samples of the entire Markov chain $\mathcal{M}$, and for each sample of $\mathcal{M}$, use Algorithm 1 to compute an estimate for CASV, potentially in parallel. These posterior samples quantify the uncertainty in CASV that arises due to the noise in data, or lack of data.

6. Numerical experiments

We now shift our focus to evaluating the performance of our framework of attribution on a large-scale real-world dataset and comparing it against competing metrics (LTA, IVH, uniform, and SV). We first discuss the dataset composed of several million real-world user paths (Section 6.1), and then the postulated Markovian model and its estimation using this real-world data (Section 6.2). Next, we present the attribution metrics (computed on data simulated from the estimated model) corresponding to various schemes along with some discussion (Section 6.3). We note that our numerics are for illustrative purposes and in particular, to see how various attribution schemes differ. Recall that attribution does depend on the underlying model (cf. discussion below Remark 2). Hence, the numbers we report in Section 6.3 are a function of the postulated model (state-action space) and its estimation in Section 6.2.

6.1. Dataset

Our real-world dataset corresponds to a single product (software) promoted and sold on the Internet by a Fortune 500 company⁹. The dataset contains several million user paths with a few hundred thousand conversions (purchases). Each path starts with a “sign-up”, i.e., the user creating an account on the company’s website. From the date of the sign-up, we have access to the user’s interaction with the company (*touchpoints*) for a period of 8 months¹⁰. The touchpoints can be classified based on the type of ad:

- **E-mail ad:** There are three related touchpoints: (1) advertiser sending an e-mail (ad action), (2) user opening an e-mail (user action), and (3) user clicking on a link in the e-mail (user action).
- **Display ad:** There are two related touchpoints: (1) advertiser showing a display ad (ad action) and (2) user clicking on a display ad (user action).

⁹ We do not disclose the name of the company and specific statistics related to real data for anonymity reasons. The company is the advertiser as opposed to a website / platform that serves ads for many advertisers.

¹⁰ Our user paths data suffers from the common issues such as we lose track of a user if she clears cookies from the web browser.

- **Paid search ad:** There is only one touchpoint corresponding to paid search impressions: user clicking on the paid search ad (user action). Due to the way data is collected, an advertiser usually does not know whether a paid search ad is shown to a user if the user does not click on the ad.

There are two additional touchpoints: (1) “sign-up” (user creates an account) and (2) “conversion” (user buys the product). If the user converted within 8 months, we truncate the path at the time of conversion (since we are interested in activity up to first conversion). The quit state is not explicitly observed in the data and we use the following rule to determine a transition to q : if there is no activity (user and advertiser) in the last 30 days and in the future, we mark the next state as q . Not every path ends in the conversion state or the quit state at the end of the 8 months period and we let it be that way (as opposed to forcing it to transition to the quit state).

We discarded certain dubious paths after discussing with our industry partner. In particular, paths with over 100 touchpoints were thrown away with the suspicion of possible bots and paths with any illogical sequence (for example, e-mail opened before being received) were also discarded (possible data recording errors). The occurrence of such paths was very low (less than 1%).

6.2. Markovian model and estimation

Our Markov chain in earlier sections is defined with abstract action and state spaces to showcase the flexibility of our framework. For the numerics, we use the following construction.

The action space $\mathbb{A}$ consists of four actions: (1) no-ad, (2) e-mail, (3) display ad, and (4) paid search *click*¹¹. In our dataset, the no-ad action is not observed explicitly and we use the following rule to “implant” no-ad actions in the user paths: if there has been no touchpoint activity (user and advertiser) in the past τ days, we implant the no-ad action. We use $\tau = 10$ in our numerics and perform sensitivity analysis with respect to τ after we present the results (at the end of Section 6.3).

We assume that that a user goes through the following states during his decision process: (1) unaware, (2) aware, (3) interest, and (4) desire. This definition of the state space $\mathbb{S}$ is motivated by the widely used conversion funnel in the marketing literature (Strong 1925, Howard and Sheth 1969, Barry 1987, Bettman et al. 1998, Court 2009, Elzinga et al. 2009, Kotler and Armstrong 2010, Mulpuru 2011, Jansen and Schuster 2011, Bruce et al. 2012). However, unlike most of this literature, we assume that our microscopic user level data allows us to objectively define the states using the touchpoints as follows:

- **Unaware:** User has received no e-mail, no display ad, and clicked on no paid search ad so far.
- **Aware:** At least one ad (e-mail or display) received by the user so far.
- **Interest:** At least one e-mail opened by the user so far.

¹¹ Since we only observe a paid search impression when it is clicked, we use the “click” as an action even though it is a user action (as opposed to an ad action). This is a limitation of the data and affects all attribution metrics, i.e., it is not specific to the framework we propose.

- **Desire:** At least one ad (e-mail, display, or paid search) clicked by the user so far.

Clearly, a “back” transition (for example, transition from aware to unaware) is impossible in our state space construction but we allow for the possibility of a “jump” (for example, transition from unaware to interest). Furthermore, transitions to quit and conversion states are possible from any of the four states in $\mathbb{S}$. (The state space can be made more granular depending on the needs of an advertiser and the type of data available. For instance, being “aware” as a result of receiving an e-mail might be different from being “aware” due to seeing a display impression. Furthermore, the “level of awareness” might depend on the number of ads seen and hence, it might be worthwhile to define multiple states capturing different levels of awareness. We kept our construction simple for illustrative purposes.)

Having defined the action and state spaces, we discuss the estimation of the corresponding Markov model $\mathcal{M}$ using real data. In particular, we need to estimate λ , β , and $\{\mathbf{P}^a\}_{a \in \mathbb{A}}$. The touchpoints in the dataset allow us to construct “ (s, a, s') ” tuples. Since each user path starts with the “sign-up” touchpoint, the initial state is always “unaware” and hence, $\lambda_1 = 1$. To estimate β_s^a for all $(s, a) \in \mathbb{S} \times \mathbb{A}$, we simply compute the empirical frequencies, i.e., we count the number of times action a was taken at state s and divide it by the number of times we observe state s in the dataset. We employ the same technique¹² to estimate $p_{ss'}^a$ for all $(s, a, s') \in \mathbb{S} \times \mathbb{A} \times \mathbb{S}^+$. As mentioned in Section 5.2, it is important to understand the impact of finite dataset on the CASV estimate. The CASV estimates reported here are robust given the large number (several million) of user paths in our dataset. We perform the following experiment to check for robustness. Assuming the estimated model to be the ground truth, we sample $D \in \{100, 1000, 10000\}$ paths. We use these D paths to estimate the model (via “empirical frequencies”), which allow us to obtain an estimate of CASV (via Algorithm 1). We numerically analyze the convergence of the CASV estimate for different values of D to the CASV under the assumed true model. Remarkably, the CASV estimates are very close to the true value for D as low as 10,000. Since the size of our real-world data is several million, we conclude that the numerical results presented here are very robust to statistical errors. The details of this experiment are in Appendix C.

We acknowledge that our simple estimation scheme could potentially be improved. In particular, there might be undesired endogeneity effects in our data, which might result in biased estimates. For instance, a user who is more inclined to convert might be less likely to receive an ad, which is against the spirit of our β -randomized policy. However, we note that such estimation issues are

¹² We also experimented with the Bayesian approach discussed at the end of Section 5.2. Due to the enormous number of user paths in our dataset, the posterior variance in the Bayesian approach was extremely small and hence, the results obtained were very similar. We do not present them to be concise.

not peculiar to CASV but they also affect other attribution schemes that require the model to be estimated (IVH for instance) and hence, we place these challenges outside the scope of this work.

The summary¹³ of the parameter estimates is as follows. In terms of actions, e-mail has the highest frequency ranging from 40% to 90% depending on the state followed by the no-ad action (10% to 50%). Combined, e-mail and no-ad actions account for over 85% of the actions at each state. The action intensity of display impression ranges from 2% to 10% and of paid search click ranges from 0.1% to 7%¹⁴. In terms of transition probabilities, we note that from each state in $\mathbb{S}$, the eventual conversion probability is positive under each action, which validates the absorption assumption (Assumption 1). Furthermore, the self-loop probability for each state-action pair (except for the no-ad action at the unaware state and the impossible self-transitions) is over 90%, indicating “slow-moving” users. For the no-ad action at the unaware state, it is around 50%. In addition, there are other patterns in the estimates that are consistent with a priori expectations. For example, a user is around 5 times more likely to move from the “aware” state to the “interest” state under the e-mail action as compared to the no-ad action. Moreover, the one-step conversion probabilities are highest for paid search click (5% to 10% depending on the state) followed by display impression (0.2% to 1.5%). As one would expect, the eventual conversion probability h_s^β increases monotonically with $s \in \mathbb{S}$, i.e., as a user goes deeper into the conversion funnel, she becomes more likely to buy the product. For instance, a user is around twice more likely to purchase if she is in the “desire” state as compared to the “unaware” state.

We also observe some estimates to be counterintuitive. For instance, our estimate of one-step conversion probability under the no-ad action is higher than under the e-mail action for all states except “unaware.” This might be a consequence of the endogeneity issue we alluded to earlier or advertising activity (such as offline ads) not captured in the online data. Recall that we use the baseline no-ad action as a proxy for all offline promotion efforts. Furthermore, for all three ad actions (e-mail, display, and paid search click), the one-step conversion probability from the “unaware” state is higher than in the “aware” state. We believe this is primarily a result of the user actually being acquainted with the product (via offline channels for instance).

6.3. Results and discussion

Given the model estimate from real data, we estimate CASV using Algorithm 1 with $N = 100,000$ (number of paths to be sampled)¹⁵. It took less than a second to compute the attribution for each

¹³ We do not report the exact numbers to protect the private data of our provider.

¹⁴ Interestingly, the highest (among all the states) action intensity of paid search click was at the first state in the funnel (“unaware”). We believe this is a result of “unaware” users actually knowing the product (from offline channels for instance).

¹⁵ We repeated our computation using different seeds and obtained almost identical results, indicating $N = 100,000$ is high enough.

scheme¹⁶ on a single core of Intel Xeon E5-2650v2 2.6 GHz processor with 8 GB of RAM. This highlights the scalability of our approach. Figures 4 and 5 display the results. Interestingly, the five attribution schemes output quite different results and we discuss them next. We first discuss the attributions to each action (Figure 4), and then, comment on the state-specific attributions to each action (Figure 5).

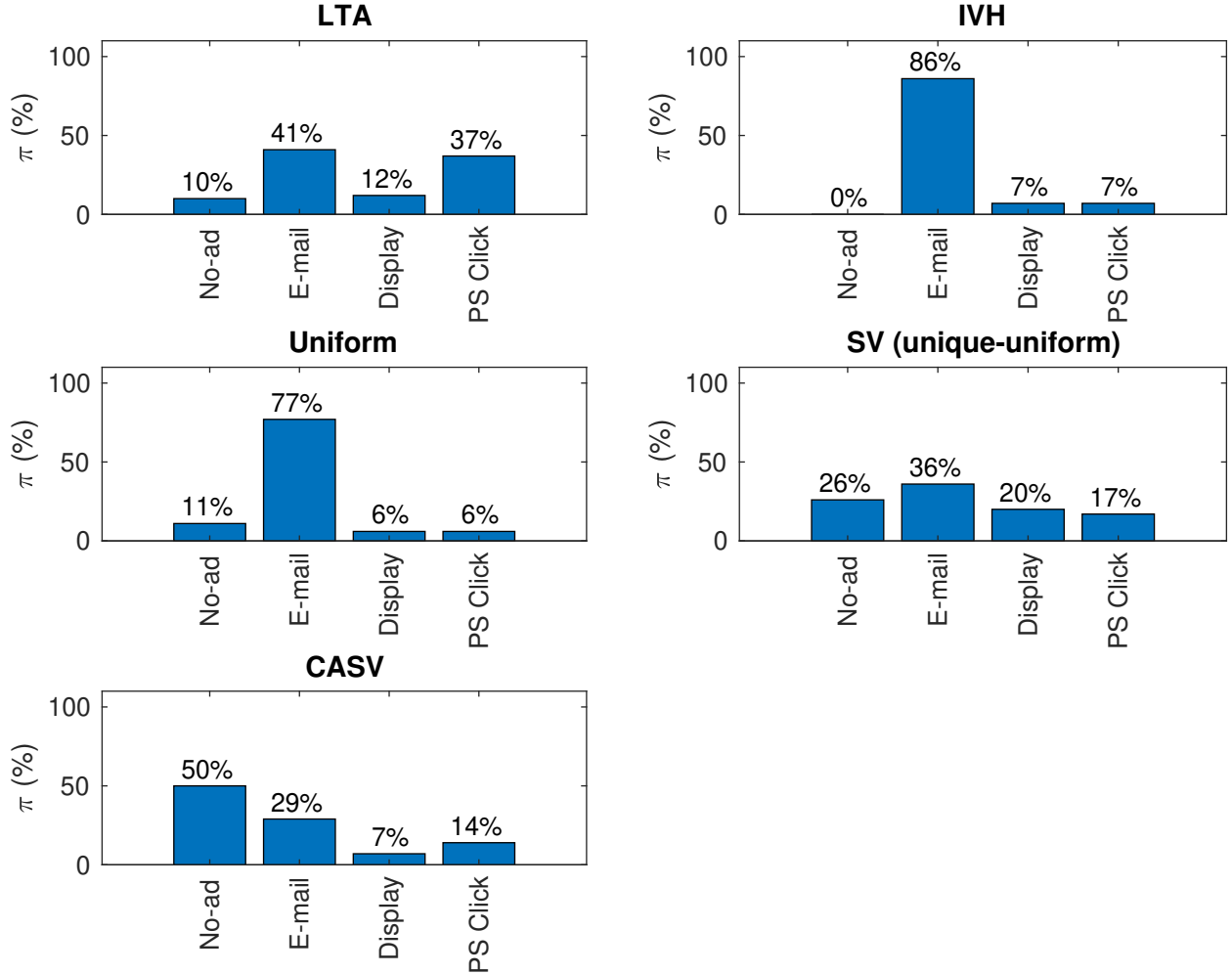


Figure 4 Attributions to different actions under various schemes with $\tau = 10$. We report the percentage attributions to each action by aggregating over states, i.e., $\sum_s \pi_s^a / \sum_{(s', a')} \pi_{s'}^{a'}$ for each $a \in \mathbb{A}$.

In terms of attributions to actions (Figure 4), LTA allocates the majority of the share to e-mail and paid search click, which seems appropriate for LTA since e-mail ads are the most prevalent in our dataset and paid search click strongly hints a higher level of customer engagement (since it is a user action as opposed to an ad action) and hence, a relatively high one-step conversion

¹⁶ For LTA, IVH, and uniform, computing attribution is straightforward (see Section 3). To estimate SV, we used the unique-uniform characterization (Proposition 1).

probability. Since LTA does not adjust for the counterfactual, the no-ad action receives little credit. IVH allocates almost all the credit to e-mail, which aligns with the facts that IVH scales with action intensity (see (1)) and e-mail action occurs around 10 times more often than display and paid search click actions. We make a similar observation for uniform attribution. Due to its unique-uniform nature, SV accounts for the fact that players corresponding to e-mail action appear multiple times in certain paths and hence, the credit allocated to e-mail shrinks to roughly half (compared to uniform). This can be seen as a correction to other schemes (IVH and uniform in particular) allocating higher credit to e-mail than it deserves simply because it appears more often in the paths. Compared with LTA, SV allocates less value to paid search click, which aligns with the observation that LTA allocates more credit than appropriate to channels appearing later in the conversion funnel. Finally, CASV corrects SV by adjusting for the counterfactual. As expected, CASV takes away a portion from e-mail, display, and paid search click and allocates it to the no-ad counterfactual. This results in no-ad receiving roughly half of the total system value¹⁷. The action that is least affected by the counterfactual adjustment is paid search click, which seems reasonable since replacing paid search click (a user action) by no-ad (an ad action) can create a big difference in total system value and hence, the counterfactual value would be small.

Next, we briefly discuss the state-action-specific attributions (Figure 5), which are reported on an absolute scale (contrast with the percentages in Figure 4). As is evident, each attribution scheme can output quite different allocations to various states for a given action. LTA allocates the most to paid search click at the “desire” state. Similar to the empirical finding of Blake et al. (2015), it is possible that a user in a state of “desire” is already inclined to buy the product and performs a Google search, leading to a paid search click and a conversion. However, in the counterfactual scenario, even if the paid search click did not occur, the user might have bought the product (by clicking on the “organic” link). In fact, the relatively low allocation to paid search click at the “desire” state under CASV provides empirical support to such a view. In terms of e-mail, existing heuristics such as LTA, IVH, and uniform have an increasing pattern in the attribution allocated with respect to the user state whereas CASV allocates the most to the first state. This also aligns with the observation above, suggesting that the existing heuristics fail to account for the counterfactual appropriately. Interestingly, all the schemes allocate a considerable amount to paid search click at the “unaware” state. This might be a result of “unaware” users actually being aware of the product (from offline sources for instance) and searching the product (on Google for example) just to make a purchase but clicking the paid search ad in the process. In theory, CASV should

¹⁷ As noted earlier (around Equation (7) and when revisiting Example 5 in Section 4.4), the attribution to no-ad factors in the value generated by advertising actions that are not modeled (offline ads for instance) and hence, is not necessarily the value attributed *only* to no-ad.

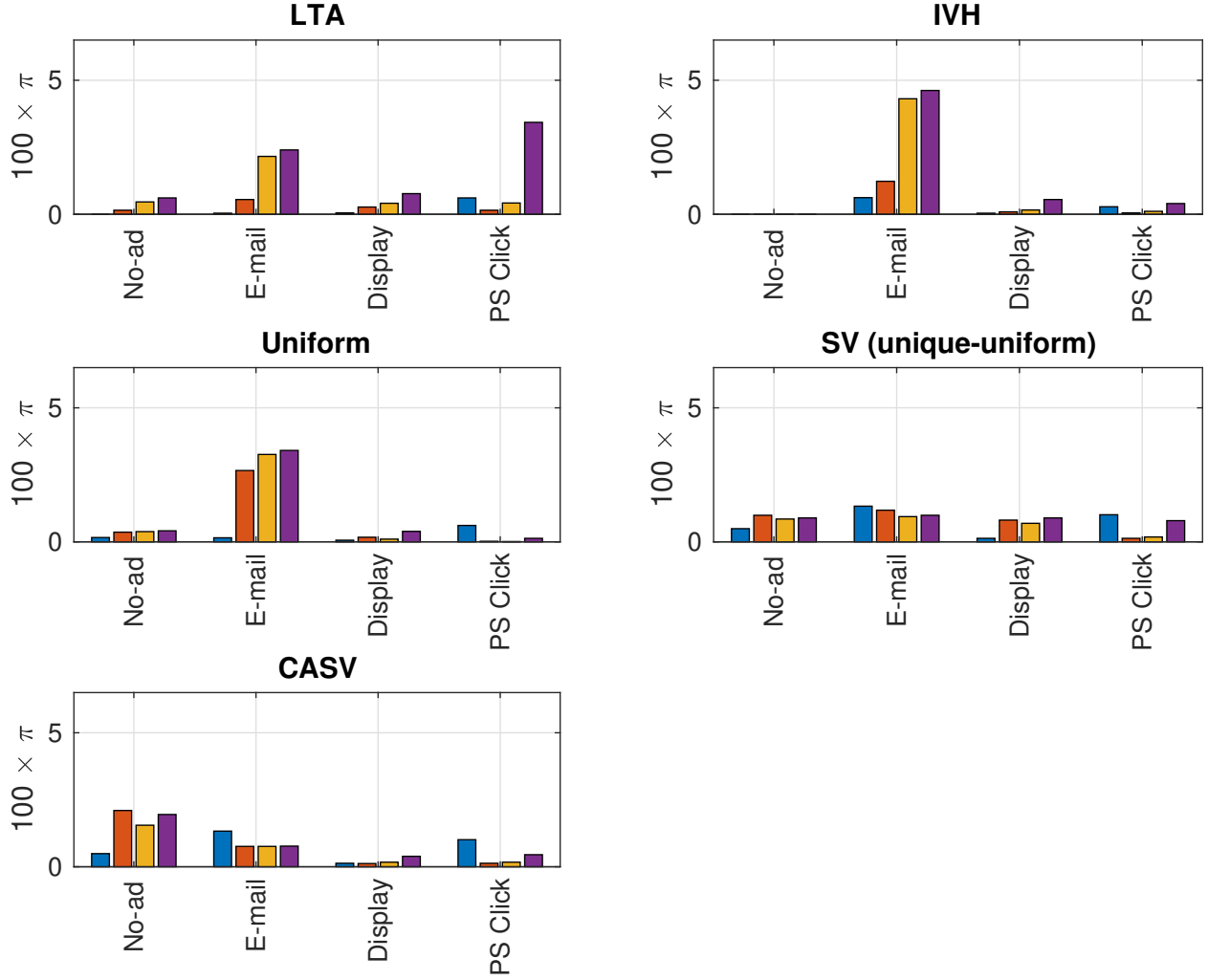


Figure 5 Attributions to state-action pairs under various schemes with $\tau = 10$. We report the attributions π_s^a for all $(s, a) \in \mathbb{S} \times \mathbb{A}$. The four colors correspond to the four states in $\mathbb{S}$, arranged in the natural order (unaware, aware, interest, desire).

adjust for this phenomenon by allocating the “offline value” to the no-ad counterfactual. However, this correction is not reflected in our numerics due to the hidden nature of data corresponding to paid search impressions (recall we only observe them when they are clicked).

Recall that π_s^a is the value attributed to the pair $(s, a) \in \mathbb{S} \times \mathbb{A}$ per sample path of the Markov chain. Another metric of interest is the *per-impression* value of an ad action, i.e., $z_s^a := \frac{\pi_s^a}{\mu_s^a \beta_s^a}$ for all $(s, a) \in \mathbb{S} \times \mathbb{A}$. In Table 1, we report $[z_s^a]_{s \in \mathbb{S}}$ for all actions other than the no-ad action. We omit $[z_s^a]_{s \in \mathbb{S}}$ for the no-ad action since the no-ad action is a mathematical construct needed for our theory – it does not correspond to a concrete action taken by the advertiser. For all attribution schemes and user states, paid search click has the highest value, followed by display, with e-mail having the lowest value. In fact, the value of a paid search click can be an order higher than the other two. Given paid search *click* is a user action and display ads are at least seen by the user

(contrast with e-mails that can remain unopened), this appears reasonable. Across all attribution schemes, ad impressions are most valuable when the user is unaware. This supports the viewpoint that advertising is more valuable in building product awareness than in the later stages of the conversion funnel, where the baseline counterfactual value can drive conversions as well.

In addition to these macro-level observations, it is worth zooming into individual attribution schemes as well. Even in the desire state, LTA places the same value (around 6 cents) on a paid search click as in the unaware state. On the other hand, the counterfactual aware CASV values the paid search click at under a cent at the desire state and over 10 cents in the unaware state. Interestingly, the widely used IVH heuristic overvalues the ad actions (w.r.t. all other schemes), which is in line with our earlier discussion regarding IVH violating budget-feasibility. This observation provides one explanation to the finding in Blake et al. (2015). Although the sample path attribution to email under the uniform attribution scheme is comparable to the other attribution schemes, the per-impression value of e-mail under uniform attribution is considerably lower because e-mail appears very frequently in the data. For example, in the unaware state, the per-impression value of e-mail under uniform attribution is 10 times lower than under unique-uniform (0.27 cents vs. 2.36 cents). Finally, the per-impression value of each ad action under CASV is never higher than that under SV, which reflects the fact that CASV “removes” the counterfactual value when valuing an ad.

We conclude this section by performing sensitivity analysis on τ . Recall that to implant the no-ad action in raw user paths, we used $\tau = 10$. We also experiment with a lower value of $\tau = 7$ (Figure 9 in Appendix D) and a higher value of $\tau = 14$ (Figure 10 in Appendix D) and analyze the resulting changes in the attribution numbers. All the discussions associated with the case of $\tau = 10$ (Figure 4) still apply qualitatively, indicating robustness of high-level insights to the choice of τ . We do see changes in the actual numbers but they align with our expectation. For example, with $\tau = 7$, we implant more instances of no-ad action (as compared to $\tau = 10$) and hence, expect it to receive more attribution (and vice versa with $\tau = 14$).

7. Value of state-specific attribution

In our cooperative game theory setup presented in Section 4, we defined each state-action pair $(s, a) \in \mathbb{S} \times \mathbb{A}$ to be a player (*state-specific model*) and hence, had CASV $\psi_s^{a, \text{Shap}}$ corresponding to each state-action pair. As we mentioned previously, a naïve implementation to *exactly* compute $\{\psi_s^{a, \text{Shap}}\}_{(s, a) \in \mathbb{S} \times \mathbb{A}}$ using (5) would have an exponential runtime in the size of $\mathbb{S} \times \mathbb{A}$. Interestingly, there is an alternative view that reduces the computational complexity of a naïve implementation to be exponential only in the size of $\mathbb{A}$, which might be tractable. In particular, one can define players in the cooperative game to be the actions in $\mathbb{A}$ (*aggregated model*) and compute CASV

		E-mail	Display	PS Click
LTA	Unaware	0.07	1.33	6.78
	Aware	0.02	0.21	5.47
	Interest	0.05	0.32	8.48
	Desire	0.12	0.33	6.53
IVH	Unaware	9.40	10.41	26.58
	Aware	0.47	0.62	16.81
	Interest	0.87	1.08	19.17
	Desire	1.90	2.06	6.51
Uniform	Unaware	0.27	1.59	6.77
	Aware	0.12	0.13	0.65
	Interest	0.08	0.08	0.25
	Desire	0.16	0.17	0.25
SV	Unaware	2.36	3.56	11.32
	Aware	0.05	0.63	4.59
	Interest	0.02	0.54	3.56
	Desire	0.05	0.39	1.50
CASV	Unaware	2.36	3.56	11.32
	Aware	0.03	0.09	4.58
	Interest	0.02	0.13	3.55
	Desire	0.04	0.17	0.85

Table 1 Per-impression values of each state-action pair under various schemes with $\tau = 10$. We report the per-impression value z_s^a (multiplied by 100) for all $(s, a) \in \mathbb{S} \times \mathbb{A}$ (except no-ad). To interpret the scale, if the sale of one unit of the product generates a value of \$1 (as stated in Section 2.3), then the reported numbers are in cents. We do not report the per-impression value of no-ad $[z_s^{\text{no-ad}}]_{s \in \mathbb{S}}$ since it is not a concrete action taken by the advertiser.

$\bar{\psi}^{a, \text{Shap}}$ for each action $a \in \mathbb{A}$. Under such a view, a coalition $\bar{\mathcal{X}}$ corresponds to a collection of actions, i.e., $\bar{\mathcal{X}} \subseteq \mathbb{A}$. To be more precise, we define the empty coalition in the same way as we defined earlier, i.e., it corresponds to the Markov chain containing all the states but action 0 being taken at all of them w.p. 1. If an action $a \in \mathbb{A}$ is added to the empty coalition, then the advertiser takes action a at each state $s \in \mathbb{S}$ w.p. β_s^a and so on. Under this aggregated model, each action $a \in \mathbb{A}$ receives an attribution of

$$\bar{\psi}^{a, \text{Shap}} := \sum_{\bar{\mathcal{X}} \subseteq \mathbb{A} \setminus \{a\}} \bar{w}_{|\bar{\mathcal{X}}|} \times \{v(\bar{\mathcal{X}} \cup \{a\}) - v(\bar{\mathcal{X}} \cup \{1^a\})\} \quad \text{where} \quad \bar{w}_{|\bar{\mathcal{X}}|} := \frac{|\bar{\mathcal{X}}|!(n - |\bar{\mathcal{X}}| - 1)!}{n!} \quad (9)$$

and 1^a denotes the counterfactual player, i.e., the no-ad action (action 1) is taken at state s w.p. β_s^a (in addition to the default intensity of β_s^1) for all $s \in \mathbb{S}$. Similar to the state-specific model, CASV in the aggregated model can be expressed as $\bar{\psi}^{a, \text{Shap}}(\mathcal{M}) = \bar{\pi}^{a, \text{Shap}}(\mathcal{M}) - \bar{\pi}^{a, \text{Shap}}(\bar{\mathcal{M}}^a)$ where

$$\bar{\pi}^{a, \text{Shap}}(\mathcal{M}) := \sum_{\bar{\mathcal{X}} \subseteq \mathbb{A} \setminus \{a\}} \bar{w}_{|\bar{\mathcal{X}}|} \times \{v_{\mathcal{M}}(\bar{\mathcal{X}} \cup \{a\}) - v_{\mathcal{M}}(\bar{\mathcal{X}})\}, \quad (10)$$

$\mathcal{M}$ denotes the original Markov chain and $\bar{\mathcal{M}}^a$ denotes the counterfactual network for a (in an aggregated sense), i.e., it is identical to $\mathcal{M}$ except that we replace the transition probabilities of

(s, a) by those of $(s, 1)$ for all $s \in \mathbb{S}$. Furthermore, SV $\bar{\pi}^{a, \text{Shap}}$ for $a \in \mathbb{A}$ in the aggregated model can be characterized as a unique-uniform scheme using the same proof technique as for Proposition 1:

$$\bar{\pi}^{a, \text{Shap}} = \mathbb{E}_{\mathcal{P} \sim \mathcal{M}} [\bar{w}^a(\mathcal{P})] \quad \text{where} \quad \bar{w}^a(\mathcal{P}) := \begin{cases} \frac{1}{\bar{u}(\mathcal{P})} & \text{if } \mathcal{P} \text{ converts and } a \in \bar{\mathcal{P}} \\ 0 & \text{otherwise.} \end{cases}$$

The function $\bar{u}(\cdot)$ returns the number of unique actions and $\bar{\mathcal{P}}$ denotes the actions in the path $\mathcal{P}$ (over state-action pairs), i.e., $\bar{\mathcal{P}} := \{a : (s, a) \in \mathcal{P}\}$. Thus, the CASV $\bar{\psi}^{a, \text{Shap}}$ for $a \in \mathbb{A}$ in the aggregated model equals

$$\bar{\psi}^{a, \text{Shap}} = \mathbb{E}_{\mathcal{P} \sim \mathcal{M}} [\bar{w}^a(\mathcal{P})] - \mathbb{E}_{\mathcal{P} \sim \bar{\mathcal{M}}^a} [\bar{w}^a(\mathcal{P})],$$

which is analogous to Theorem 2.

If the size of $\mathbb{A}$ is relatively small (which might be the case for certain retailers), CASV can be computed *exactly* in the aggregated model using (9). Accordingly, it is of interest to check whether using the granular state-specific model (which is computationally demanding) provides some quantifiable value over the aggregated model. However, one needs to be careful when comparing the attributions of two models as they are on different “scales.” The state-specific model outputs attribution at a state-action level whereas the aggregated model outputs at an action level. Furthermore, decomposing the action level allocation of the aggregated model to the state-action level seems non-trivial. On the other hand, aggregating the state-action level allocations of the state-specific model is straightforward ($\sum_{s \in \mathbb{S}} \psi_s^{a, \text{Shap}}$). Therefore, we will compare $\sum_{s \in \mathbb{S}} \psi_s^{a, \text{Shap}}$ to $\bar{\psi}^{a, \text{Shap}}$ to answer the following question: does computing state-specific attribution and then aggregating over states provide a different answer than directly computing attribution that is not state-specific? Clearly, this is a stronger notion of comparison in the sense that if $\sum_{s \in \mathbb{S}} \psi_s^{a, \text{Shap}}$ differs from $\bar{\psi}^{a, \text{Shap}}$, then any state-specific decomposition of $\bar{\psi}^{a, \text{Shap}}$ would be “inappropriate.”

Our unique-uniform characterizations imply that

$$\sum_{s \in \mathbb{S}} \psi_s^{a, \text{Shap}} - \bar{\psi}^{a, \text{Shap}} = \mathbb{E}_{\mathcal{P} \sim \mathcal{M}} \left[\sum_{s \in \mathbb{S}} w_s^a(\mathcal{P}) - \bar{w}^a(\mathcal{P}) \right] - \left[\sum_{s \in \mathbb{S}} \mathbb{E}_{\mathcal{P} \sim \mathcal{M}_{(s, a)}} [w_s^a(\mathcal{P})] - \mathbb{E}_{\mathcal{P} \sim \bar{\mathcal{M}}^a} [\bar{w}^a(\mathcal{P})] \right],$$

which gives a path-based view of the difference between the two approaches. For instance, in a network with zero counterfactual value, if a path contains x unique state-action pairs with $x - 1$ of them corresponding to the same action, the state-specific weight function allocates $1/x$ to each state-action pair and hence, the repeated action receives an attribution of $(x - 1)/x$ to account for its state-dependent impact. On the contrary, the aggregated model allocates $1/2$ to both the unique actions. Though useful, such a path-based view does not build an understanding in terms of the type of interactions among players captured (and missed) by the two approaches. To facilitate such a fundamental understanding, we define two types of coalitions (assuming the underlying players are state-action pairs).

Type 1 coalition. In a type 1 coalition (say $\mathcal{X}_1$), if a player (s, a) is in $\mathcal{X}_1$, then the player (s', a) is in $\mathcal{X}_1$ for all $s' \in \mathbb{S}$. In other words, if an action is present at a single state, then it is present at all the states. We denote by $\mathbb{T}_1$ the set of all type 1 coalitions. Type 1 coalitions correspond to the coalitions in the aggregated model. Intuitively, both the aggregated and the state-specific models capture the interactions present in type 1 coalitions. However, it is not clear whether the “weights” assigned to each of these interactions are the same in the two models. (We shed more light on this issue in Theorem 3.)

Type 2 coalition. Any coalition that is a subset of $\mathbb{S} \times \mathbb{A}$ but not a type 1 coalition is defined as a type 2 coalition. In other words, type 2 coalitions have at least one action that is present at one or more states but not at all the states. We define $\mathbb{T}_2$ as the collection of all type 2 coalitions. It should be intuitively clear from (9) that the aggregated model will miss all the interactions present in type 2 coalitions. (We formalize this intuition in Theorem 3.)

We are now equipped to precisely quantify the value of state-specific attribution in terms of the interactions among the underlying players and we do so in Theorem 3, the proof of which is presented in Appendix E.

Theorem 3 (Value of state-specific attribution). *For any action, computing state-specific CASV and then aggregating over states is not the same as computing CASV over the aggregated model. In particular, for all $a \in \mathbb{A}$,*

$$\sum_{s \in \mathbb{S}} \psi_s^{a, \text{Shap}} - \bar{\psi}^{a, \text{Shap}} = \sum_{\mathcal{X} \in \mathbb{T}_1} \{c^a(\mathcal{X}) - \bar{c}^a(\mathcal{X})\} \{v_{\mathcal{M}}(\mathcal{X}) - v_{\mathcal{M}^a}(\mathcal{X})\} + \sum_{\mathcal{X} \in \mathbb{T}_2} \sum_{s \in \mathbb{S}} c_s^a(\mathcal{X}) \{v_{\mathcal{M}}(\mathcal{X}) - v_{\mathcal{M}(s,a)}(\mathcal{X})\},$$

where $c^a(\mathcal{X}) := \sum_{s \in \mathbb{S}} c_s^a(\mathcal{X})$,

$$c_s^a(\mathcal{X}) := \begin{cases} w_{|\mathcal{X}|-1} & \text{if } (s, a) \in \mathcal{X} \\ -w_{|\mathcal{X}|} & \text{otherwise} \end{cases} \quad \text{and} \quad \bar{c}^a(\mathcal{X}) := \begin{cases} \bar{w}_{|\bar{\mathcal{X}}|-1} & \text{if } a \in \bar{\mathcal{X}} \\ -\bar{w}_{|\bar{\mathcal{X}}|} & \text{otherwise.} \end{cases}$$

The notation $\bar{\mathcal{X}}$ denotes the actions in $\mathcal{X}$, i.e., $\bar{\mathcal{X}} := \{a : (s, a) \in \mathcal{X}\}$.

We now parse the mathematical result presented above. The first term in the difference corresponds to type 1 coalitions. Such coalitions are captured by both the models. However, it is not true that the two models assign equal weights to them ($c^a(\mathcal{X})$ vs. $\bar{c}^a(\mathcal{X})$). On the other hand, the second term corresponds to type 2 coalitions, which the aggregated model completely disregards (note that there is no $\bar{c}$ weight in the second term). Hence, the aggregated model misses a multitude of interactions that are captured by the state-specific model.

Remark 3. It is easy to show that $c^a(\mathcal{X})$ for any coalition $\mathcal{X} \subseteq \mathbb{S} \times \mathbb{A}$ and action $a \in \mathbb{A}$ equals

$$c^a(\mathcal{X}) = m^a(\mathcal{X})w_{|\mathcal{X}|-1} - (m - m^a(\mathcal{X}))w_{|\mathcal{X}|},$$

where the scalar $m^a(\mathcal{X})$ equals the number of players in the set $\{(s, a)\}_{s \in \mathbb{S}}$ that exist in the coalition $\mathcal{X}$ and m equals the number of states in $\mathbb{S}$ (refer to the definition of $\mathbb{S}$ in Section 2.1). Furthermore, for $\mathcal{X} \in \mathbb{T}_1$, we have

$$m^a(\mathcal{X}) = \begin{cases} m & \text{if } a \in \bar{\mathcal{X}} \\ 0 & \text{otherwise.} \end{cases}$$

As a result, the coefficient in the first term simplifies to

$$c^a(\mathcal{X}) - \bar{c}^a(\mathcal{X}) = \begin{cases} mw_{|\mathcal{X}|-1} - \bar{w}_{|\bar{\mathcal{X}}|-1} & \text{if } a \in \bar{\mathcal{X}} \\ \bar{w}_{|\bar{\mathcal{X}}|} - mw_{|\mathcal{X}|} & \text{otherwise.} \end{cases}$$

Having quantified the difference between the state-specific and aggregated models, we show that the ratio of the two models can be arbitrarily large, which accentuates the importance of state-specific attribution.

Proposition 2. *The ratio of the aggregated model's attribution to the aggregation of the state-specific model's attribution can be arbitrarily large. Mathematically,*

$$\sup_{\mathcal{M}} \sup_{a \in \mathbb{A}} \frac{\bar{\psi}^{a, \text{Shap}}(\mathcal{M})}{\sum_{s \in \mathbb{S}} \psi_s^{a, \text{Shap}}(\mathcal{M})} = \infty.$$

A proof of Proposition 2 is presented in Appendix E. We conclude this section by showing Figure 6, which displays the difference between the two models (state-specific and aggregated) on the simulated dataset from Section 6. The aggregated model underestimates the contributions of no-ad and e-mail actions by around 10%.

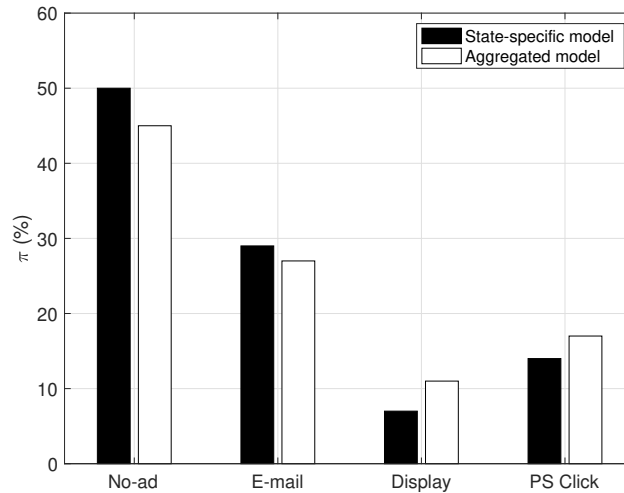


Figure 6 Value of state-specific attribution on the simulated dataset. For the state-specific model, we report the percentage attributions to each action by aggregating over states, i.e., $\sum_s \psi_s^{a, \text{Shap}} / \sum_{(s', a')} \psi_{s'}^{a', \text{Shap}}$ for each $a \in \mathbb{A}$. For the aggregated model, we report $\bar{\psi}^{a, \text{Shap}}$ for each $a \in \mathbb{A}$.

8. Conclusions and further research

In this paper, using a Markovian model for user behavior, we propose counterfactual adjusted Shapley value (CASV) as a new metric for attribution in online advertising. We establish its theoretical foundations and appropriateness in two ways. First, we show the robustness of the proposed measure over various canonical settings in which the existing metrics fail. Second, we provide an underlying axiomatic framework motivated by game theory and causality that supports our choice. We show that the CASV in our Markovian model is an adjustment to the unique-uniform attribution scheme. Finally, we propose multiple approaches to estimate our metric for the Markov chain model and show its scalability through numerical experiments on a real-world large-scale dataset, in addition to benchmarking it against existing metrics.

Our work proposes a very general Markovian model for the customer behavior and shows that the CASV metric can be efficiently computed for any such Markov chain. In Section 6, we propose a state space definition that is adequate for the particular application. Since the resulting attribution depends on the definition of state, defining an appropriate state space is important. However, we expect that the appropriate definition for the state is likely to be very context specific. Learning context-specific state aggregation and Markov chain dynamics from data is an active area of research. We refer the reader to Hallak et al. (2013) and references therein where the authors posit a model selection procedure when the candidate models are generated by domain experts. As noted below Remark 2, robustness of the existing attribution schemes to model perturbations is not well understood. An interesting direction would be to develop model-robust attribution schemes. A key challenge here is to formalize the notion of two models being “close.” Relatedly, a possible approach to model selection in such contexts may be to map the data to a (unique) “minimal” state-space representation, which is a topic of recent interest (see, e.g., Bennouna et al. (2021), Zhu et al. (2020), and Ni et al. (2021)). Furthermore, since our axiomatic justification does not require the Markovian assumption, characterizing CASV under non-Markovian models is of interest.

Our work does not account for the cost of ad actions; a cost-aware attribution scheme is an interesting extension to explore since a relatively cheap but less effective ad action might be more valuable than an expensive but more effective ad action. Further, in this work, we implicitly assume that the various ad actions / channels cooperate with each other towards the common goal of maximizing system value instead of being strategic and maximizing their individual values. Motivated by the works of Berman (2018) and Abhishek et al. (2017), it would be interesting to analyze our framework when the individual channels are strategic. However, it is unclear if a *cooperative* game theory framework is even appropriate under this strategic setting. For example, the *unique*-uniform scheme provides each channel an incentive to pose as two smaller but distinct channels.

Our framework could be of interest in other domains. For instance, there has been a growing interest in the field of “interpretable machine learning”, where the goal is to attribute the output of a prediction model to the individual features of the input (Baehrens et al. 2010, Sundararajan et al. 2017, Lundberg and Lee 2017) or to the components of the prediction model (Dhamdhere et al. 2018, Leino et al. 2018). Extensions of the ideas presented in this work have implications to answering such questions.

In this work, we modify Shapley value using Rubin’s definition of causality (Rubin 1974) and it is of interest to explore alternative formulations for causality. See, e.g. Pearl (2009), Halpern and Pearl (2005), Chockler and Halpern (2004), Hitchcock (1997), Morgan and Winship (2014), Collins et al. (2004), Eells (1991) and Hume (2003) for approaches to causality in the computer science and philosophy literature. Chockler and Halpern (2004) proposed metrics such as “degree of responsibility” and “blame” in order to *quantify* the causal effect of one variable on another. However, computing these metrics is computationally expensive. Furthermore, these metrics lack axiomatic support. Also, similar to IVH, these metrics are not budget-balanced. In fact, Chockler and Halpern (2004) mention adjusting their notions using Shapley value as a potential future work. Accordingly, though we have unified the existing attribution literature in this work, there is more to explore in terms of the set of counterfactuals to consider.

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Appendix A: Coalition value

A key ingredient to use the concept of Shapley value is to define the value of a coalition, i.e., the value that a subset of players generates. The key decision here is with regards to the players that are *not* in the coalition, and this decision can be very context dependent.

We illustrate the issues with a simple example. An organization consists of $\mathbb{P}$ players, and each player s takes the baseline action $a = 1$, or an enhanced action $a = 2$. Let $\mathbf{a} = [a_s]_{s \in \mathbb{P}}$ denote the action of all the players. Let $f(\mathbf{a}_S, S)$ denote the value generated by the organization when only $S \subseteq \mathbb{P}$ players are present, and they take actions $\mathbf{a}_S$, with $f(\mathbf{a}_\emptyset, \emptyset) = 0$, and $f(\mathbf{1}, \mathbb{P}) > 0$, i.e., there is value to the organization even if all players take the baseline action. Next, consider a coalition $\mathcal{X} \subseteq \mathbb{P}$ of players. We have two options for treating the players $\mathbb{P} \setminus \mathcal{X}$:

1. A player $p \notin \mathcal{X}$ is treated as absent from the organization. Thus, the value $v(\mathcal{X})$ is only generated by the players in $\mathcal{X}$, i.e., $v(\mathcal{X}) = f(\mathbf{a}_\mathcal{X}, \mathcal{X})$. This is the typical specifications in SV applications. In this setting, $v(\emptyset) = 0$ since, with no players, there is no organization, and therefore, no value.

2. A player $p \notin \mathcal{X}$ is assumed to be still present in the organization and taking the baseline action $a = 1$, i.e., $v(\mathcal{X}) = f((\mathbf{a}_\mathcal{X}, \mathbf{1}_{\mathbb{P} \setminus \mathcal{X}}), \mathbb{P})$. In this case, $v(\emptyset) = f(\mathbf{1}, \mathbb{P}) > 0$, the value generated when all players in the organization take the baseline action $a = 1$. Next, consider any player $s \in \mathbb{P}$ that takes the baseline action $a_s = 1$, then $v(\mathcal{X} \cup \{s\}) = v(\mathcal{X}) = f((\mathbf{a}_\mathcal{X}, \mathbf{1}_{\mathbb{P} \setminus \mathcal{X}}), \mathbb{P})$; consequently, the SV allocated to any such player is, identically, zero. Thus, all of the additional value $f(\mathbf{a}, \mathbb{P}) - f(\mathbf{1}, \mathbb{P})$ is allocated to players taking action 2, i.e., players taking action 2 are able to “free ride” on the effort of the players taking the baseline action. The players taking the baseline action $a = 1$ at most get a share of the baseline value $f(\mathbf{1}, \mathbb{P})$, which can be arbitrarily small. Note that this approach implicitly attributes the counterfactual value of taking $a = 2$; however, it attributes action $a = 2$ too much of the value. To further highlight the “unfairness” of this free-riding, consider a setting $f(\mathbf{1}, \mathbb{P}) = 0$, i.e., the value is zero if *all* players take the baseline action, and, in addition, player s_0 is essential for the organization, i.e., $f(\mathbf{a}_\mathcal{X}, \mathcal{X}) = 0$ if $s_0 \notin \mathcal{X}$. Suppose, the player s_0 plays the baseline action $a = 1$, then s_0 gets zero attribution even though she is essential.

This simple setting shows that if there is “value” in the baseline action, and we want to be able to attribute to the baseline action, we must define coalitions by ensuring that players $\mathbb{P} \setminus \mathcal{X}$ are not contributing to value generation. In the online advertising setting, we can achieve this by assuming that the traffic exits the system with probability 1 when one employs a state-action pair $(s, a) \notin \mathcal{X}$.

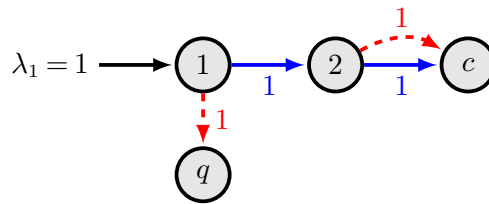


Figure 7 The action space consists of two actions: no-ad action ($a = 1$) and ad action ($a = 2$). Solid blue (dashed red) lines denote transitions if the ad action (no-ad action) is taken. The advertiser takes the ad action at state 1 w.p. 1 and takes the no-ad action in state 2 w.p. 1, i.e., $\beta_1^2 = \beta_2^1 = 1$.

To illustrate the implications of the two choices in a Markov chain setting, consider the network in Figure 7. The total value in the system is 1. States 1 and 2 are both essential for completion; however, in state 2 it does not matter whether we take the baseline no-ad action or the ad action. Thus, any reasonable attribution scheme should assign a value of $1/2$ to each of the states, and further assign a counterfactual adjusted value of $1/2$ to the ad action in state 1, and a counterfactual adjusted value of 0 to the ad action in state 2.

Under the construct in the setting 2 where the “empty” coalition consists of taking the baseline “no-ad” in each state, the ad action at state 1 receives the entire value, i.e., the baseline action in state 2 does not receive any attribution.

Appendix B: Proof of Theorem 1

Theorem 1 (Axioms). Fix a model $\mathcal{M}$. Then, CASV is the unique solution of the following four counterfactual axioms.

1. **Counterfactual efficiency:** The sum of CASVs equals the additional value generated over the counterfactual value, i.e.,

$$\sum_{(s,a) \in \mathbb{S} \times \mathbb{A}} \psi_s^{a,Shap} = v(\mathbb{S} \times \mathbb{A}) - \sum_{(s,a) \in \mathbb{S} \times \mathbb{A}} \pi_s^{a,Shap}(\mathcal{M}_{(s,a)}).$$

2. **Counterfactual symmetry:** If $(s,a) \in \mathbb{S} \times \mathbb{A}$ and $(s',a') \in \mathbb{S} \times \mathbb{A}$ are counterfactual equivalent, i.e.,

$$(A) \ v(\mathcal{X} \cup \{(s,a)\}) - v(\mathcal{X} \cup \{(s,1)^a\}) = v(\mathcal{X} \cup \{(s',a')\}) - v(\mathcal{X} \cup \{(s',1)^{a'}\}) \text{ and}$$

$$(B) \ v(\mathcal{X} \cup \{(s,1)^a, (s',a')\}) = v(\mathcal{X} \cup \{(s',1)^{a'}, (s,a)\})$$

for all $\mathcal{X} \subseteq \{\mathbb{S} \times \mathbb{A}\} \setminus \{(s,a), (s',a')\}$, then

$$\psi_s^{a,Shap} = \psi_{s'}^{a',Shap}.$$

3. **Linearity:** Consider two characteristic functions $v_1(\cdot)$ and $v_2(\cdot)$. For all $(s,a) \in \mathbb{S} \times \mathbb{A}$, we have

$$\psi_s^{a,Shap}(v_1 + v_2) = \psi_s^{a,Shap}(v_1) + \psi_s^{a,Shap}(v_2)$$

and for all $\alpha \in \mathbb{R}$,

$$\psi_s^{a,Shap}(\alpha v_1) = \alpha \psi_s^{a,Shap}(v_1).$$

4. **Counterfactual null player:** Consider a player $(s,a) \in \mathbb{S} \times \mathbb{A}$ that has a zero value-add to all coalitions that do not contain (s,a) , i.e., for all $\mathcal{X} \subseteq \{\mathbb{S} \times \mathbb{A}\} \setminus \{(s,a)\}$,

$$v(\mathcal{X} \cup \{(s,a)\}) = v(\mathcal{X} \cup \{(s,1)^a\}).$$

Then, $\psi_s^{a,Shap} = 0$.

Proof. Counterfactual efficiency and linearity follow from (6) when used with the efficiency and linearity of SV, respectively. Counterfactual null player follows from (5). For counterfactual symmetry, consider $(s,a) \in \mathbb{S} \times \mathbb{A}$ and $(s',a') \in \mathbb{S} \times \mathbb{A}$ satisfying (A) and (B) and observe that

$$\begin{aligned} \psi_s^{a,Shap} &= \sum_{\mathcal{X} \subseteq \{\mathbb{S} \times \mathbb{A}\} \setminus \{(s,a)\}} w_{|\mathcal{X}|} \times \{v(\mathcal{X} \cup \{(s,a)\}) - v(\mathcal{X} \cup \{(s,1)^a\})\} \\ &= \sum_{\mathcal{X} \subseteq \{\mathbb{S} \times \mathbb{A}\} \setminus \{(s,a), (s',a')\}} w_{|\mathcal{X}|} \times \{v(\mathcal{X} \cup \{(s,a)\}) - v(\mathcal{X} \cup \{(s,1)^a\})\} \\ &\quad + \sum_{\mathcal{X} \subseteq \{\mathbb{S} \times \mathbb{A}\} \setminus \{(s,a), (s',a')\}} w_{|\mathcal{X}|+1} \times \{v(\mathcal{X} \cup \{(s,a), (s',a')\}) - v(\mathcal{X} \cup \{(s,1)^a, (s',a')\})\} \\ &= \sum_{\mathcal{X} \subseteq \{\mathbb{S} \times \mathbb{A}\} \setminus \{(s,a), (s',a')\}} w_{|\mathcal{X}|} \times \{v(\mathcal{X} \cup \{(s',a')\}) - v(\mathcal{X} \cup \{(s',1)^{a'}\})\} \\ &\quad + \sum_{\mathcal{X} \subseteq \{\mathbb{S} \times \mathbb{A}\} \setminus \{(s,a), (s',a')\}} w_{|\mathcal{X}|+1} \times \{v(\mathcal{X} \cup \{(s,a), (s',a')\}) - v(\mathcal{X} \cup \{(s',1)^{a'}, (s,a)\})\} \\ &= \sum_{\mathcal{X} \subseteq \{\mathbb{S} \times \mathbb{A}\} \setminus \{(s',a')\}} w_{|\mathcal{X}|} \times \{v(\mathcal{X} \cup \{(s',a')\}) - v(\mathcal{X} \cup \{(s',1)^{a'}\})\} \\ &= \psi_{s'}^{a',Shap}. \end{aligned}$$

To show uniqueness, consider $\{\psi_s^a\}_{(s,a) \in \mathbb{S} \times \mathbb{A}}$ such that it satisfies the counterfactual axioms. Motivated by (6), express ψ_s^a as

$$\psi_s^a = \underbrace{\psi_s^a + \pi_s^{a, \text{Shap}}(\mathcal{M}_{(s,a)})}_{=: \pi_s^a} - \pi_s^{a, \text{Shap}}(\mathcal{M}_{(s,a)})$$

for all $(s, a) \in \mathbb{S} \times \mathbb{A}$. To show $\psi_s^a = \pi_s^{a, \text{Shap}}$ for all $(s, a) \in \mathbb{S} \times \mathbb{A}$, it suffices to show that $\pi_s^a = \pi_s^{a, \text{Shap}}$ for all $(s, a) \in \mathbb{S} \times \mathbb{A}$. We do so by proving $\{\pi_s^a\}_{(s,a) \in \mathbb{S} \times \mathbb{A}}$ satisfies the four desirable properties of SV from Section 4.1 (recall $\{\pi_s^{a, \text{Shap}}\}_{(s,a) \in \mathbb{S} \times \mathbb{A}}$ is the unique solution to those properties).

Efficiency: Since $\pi_s^a = \psi_s^a + \pi_s^{a, \text{Shap}}(\mathcal{M}_{(s,a)})$ for all $(s, a) \in \mathbb{S} \times \mathbb{A}$ and $\{\psi_s^a\}_{(s,a) \in \mathbb{S} \times \mathbb{A}}$ satisfies counterfactual efficiency, we get

$$\sum_{(s,a) \in \mathbb{S} \times \mathbb{A}} \pi_s^a = \sum_{(s,a) \in \mathbb{S} \times \mathbb{A}} \psi_s^a + \sum_{(s,a) \in \mathbb{S} \times \mathbb{A}} \pi_s^{a, \text{Shap}}(\mathcal{M}_{(s,a)}) = v(\mathbb{S} \times \mathbb{A}).$$

Symmetry: Consider $(s, a) \in \mathbb{S} \times \mathbb{A}$ and $(s', a') \in \mathbb{S} \times \mathbb{A}$. We need to show that

$$\underbrace{v(\mathcal{X} \cup \{(s, a)\}) = v(\mathcal{X} \cup \{(s', a')\}) \quad \forall \mathcal{X} \subseteq \{\mathbb{S} \times \mathbb{A}\} \setminus \{(s, a), (s', a')\}}_{(\diamond)} \implies \pi_s^a = \pi_{s'}^{a'}.$$

We can use the fact that $\{\psi_s^a, \psi_{s'}^{a'}\}$ satisfy counterfactual symmetry, i.e., if (A) and (B) hold for all $\mathcal{X} \subseteq \{\mathbb{S} \times \mathbb{A}\} \setminus \{(s, a), (s', a')\}$, then

$$\pi_s^a - \pi_s^{a, \text{Shap}}(\mathcal{M}_{(s,a)}) = \pi_{s'}^{a'} - \pi_{s'}^{a', \text{Shap}}(\mathcal{M}_{(s', a')}).$$

For the purposes of a contradiction, suppose that $(\diamond)$ holds but $\pi_s^a \neq \pi_{s'}^{a'}$. It suffices to show that this results in $\{\psi_s^a, \psi_{s'}^{a'}\}$ violating counterfactual symmetry. Suppose (A) and (B) hold. Given $(\diamond)$, statement (A) implies

$$v(\mathcal{X} \cup \{(s, 1)^a\}) = v(\mathcal{X} \cup \{(s', 1)^{a'}\}) \quad \forall \mathcal{X} \subseteq \{\mathbb{S} \times \mathbb{A}\} \setminus \{(s, a), (s', a')\}.$$

Together with (B), this implies $\pi_s^{a, \text{Shap}}(\mathcal{M}_{(s,a)}) = \pi_{s'}^{a', \text{Shap}}(\mathcal{M}_{(s', a')})$:

$$\begin{aligned} \pi_s^{a, \text{Shap}}(\mathcal{M}_{(s,a)}) &= \sum_{\mathcal{X} \subseteq \{\mathbb{S} \times \mathbb{A}\} \setminus \{(s,a)\}} w_{|\mathcal{X}|} \times \{v(\mathcal{X} \cup \{(s, 1)^a\}) - v(\mathcal{X})\} \\ &= \sum_{\mathcal{X} \subseteq \{\mathbb{S} \times \mathbb{A}\} \setminus \{(s,a), (s', a')\}} w_{|\mathcal{X}|} \times \{v(\mathcal{X} \cup \{(s, 1)^a\}) - v(\mathcal{X})\} \\ &\quad + \sum_{\mathcal{X} \subseteq \{\mathbb{S} \times \mathbb{A}\} \setminus \{(s,a), (s', a')\}} w_{|\mathcal{X}|+1} \times \{v(\mathcal{X} \cup \{(s, 1)^a, (s', a')\}) - v(\mathcal{X} \cup \{(s', a')\})\} \\ &= \sum_{\mathcal{X} \subseteq \{\mathbb{S} \times \mathbb{A}\} \setminus \{(s,a), (s', a')\}} w_{|\mathcal{X}|} \times \{v(\mathcal{X} \cup \{(s', 1)^{a'}\}) - v(\mathcal{X})\} \\ &\quad + \sum_{\mathcal{X} \subseteq \{\mathbb{S} \times \mathbb{A}\} \setminus \{(s,a), (s', a')\}} w_{|\mathcal{X}|+1} \times \{v(\mathcal{X} \cup \{(s', 1)^{a'}, (s, a)\}) - v(\mathcal{X} \cup \{(s, a)\})\} \\ &= \sum_{\mathcal{X} \subseteq \{\mathbb{S} \times \mathbb{A}\} \setminus \{(s', a')\}} w_{|\mathcal{X}|} \times \{v(\mathcal{X} \cup \{(s', 1)^{a'}\}) - v(\mathcal{X})\} \\ &= \pi_{s'}^{a', \text{Shap}}(\mathcal{M}_{(s', a')}). \end{aligned}$$

Hence, we require $\pi_s^a = \pi_{s'}^{a'}$ for $\{\psi_s^a, \psi_{s'}^{a'}\}$ to satisfy counterfactual symmetry, which contradicts $\pi_s^a \neq \pi_{s'}^{a'}$.

Linearity: Since $\pi_s^a = \psi_s^a + \pi_s^{a, \text{Shap}}(\mathcal{M}_{(s,a)})$ for all $(s, a) \in \mathbb{S} \times \mathbb{A}$ and both ψ_s^a and $\pi_s^{a, \text{Shap}}(\mathcal{M}_{(s,a)})$ satisfy linearity, it follows that π_s^a does so too.

Null player: Consider $(s, a) \in \mathbb{S} \times \mathbb{A}$. We need to show that

$$\underbrace{v(\mathcal{X} \cup \{(s, a)\}) = v(\mathcal{X}) \quad \forall \mathcal{X} \subseteq \{\mathbb{S} \times \mathbb{A}\} \setminus \{(s, a)\}}_{(*)} \implies \pi_s^a = 0.$$

We can use the fact that ψ_s^a satisfies counterfactual null player, i.e.,

$$\underbrace{v(\mathcal{X} \cup \{(s, a)\}) = v(\mathcal{X} \cup \{(s, 1)^a\}) \quad \forall \mathcal{X} \subseteq \{\mathbb{S} \times \mathbb{A}\} \setminus \{(s, a)\}}_{(\square)} \implies \pi_s^a = \pi_s^{a, \text{Shap}}(\mathcal{M}_{(s, a)}).$$

For the purposes of a contradiction, suppose that $(*)$ holds but $\pi_s^a \neq 0$. It suffices to show that this results in ψ_s^a violating counterfactual null player. Suppose $(\square)$ holds. Subtract $v(\mathcal{X})$ from both sides of $(\square)$ to get

$$v(\mathcal{X} \cup \{(s, a)\}) - v(\mathcal{X}) = v(\mathcal{X} \cup \{(s, 1)^a\}) - v(\mathcal{X}) \quad \forall \mathcal{X} \subseteq \{\mathbb{S} \times \mathbb{A}\} \setminus \{(s, a)\},$$

which combined with $(*)$ yields

$$0 = v(\mathcal{X} \cup \{(s, 1)^a\}) - v(\mathcal{X}) \quad \forall \mathcal{X} \subseteq \{\mathbb{S} \times \mathbb{A}\} \setminus \{(s, a)\}.$$

This implies $\pi_s^{a, \text{Shap}}(\mathcal{M}_{(s, a)}) = 0$ and hence, we require $\pi_s^a = 0$ for ψ_s^a to satisfy counterfactual null player, which contradicts $\pi_s^a \neq 0$.

This completes the proof. □

Appendix C: Sensitivity of CASV due to finite size of data

In this appendix, we discuss the sensitivity of CASV due to the finite amount of data one observes in practice. Throughout the discussion, we assume there exists a true underlying Markov model $\mathcal{M}$ but we only observe a finite dataset of D sample paths¹⁸ $\mathcal{D} := \{\mathcal{P}_i\}_{i=1}^D$, where each path is sampled independently from $\mathcal{M}$. The goal is to estimate CASV using the data $\mathcal{D}$ and understand the difference between the estimated CASV and the true CASV.

We use the data $\mathcal{D}$ to compute an estimate $\widehat{\mathcal{M}}$ for the Markov model $\mathcal{M}$ (step 1) and then, use $\widehat{\mathcal{M}}$ to estimate CASV (step 2). We followed this process for the results presented in Section 6. For step 2, we use Algorithm 1 with the inputs $\widehat{\mathcal{M}}$ (estimated model) and $N = 100,000$ (number of Monte Carlo samples). As discussed in Section 5.2, $N = 100,000$ is large enough to obtain a stable estimate of CASV and hence, we do not discuss Monte Carlo error here. For ease of notation, we use $\{\psi_s^{a,\text{Shap}}(\mathcal{M})\}_{(s,a) \in \mathbb{S} \times \mathbb{A}}$ and $\{\psi_s^{a,\text{Shap}}(\widehat{\mathcal{M}})\}_{(s,a) \in \mathbb{S} \times \mathbb{A}}$ to denote the output of Algorithm 1 when the input model to the algorithm is $\mathcal{M}$ and $\widehat{\mathcal{M}}$, respectively¹⁹.

In order to understand the difference between the true CASV $\{\psi_s^{a,\text{Shap}}(\mathcal{M})\}_{(s,a) \in \mathbb{S} \times \mathbb{A}}$ and the estimated CASV $\{\psi_s^{a,\text{Shap}}(\widehat{\mathcal{M}})\}_{(s,a) \in \mathbb{S} \times \mathbb{A}}$, we perform the following simple yet informative experiment. We fix $\mathcal{M}$ to be the model we estimated using real-world data in Section 6 and then repeat the following steps 1,000 times:

1. Sample a dataset $\mathcal{D} := \{\mathcal{P}_i\}_{i=1}^D$ of D independent paths from $\mathcal{M}$.
2. Use $\mathcal{D}$ to obtain the estimated Markov model $\widehat{\mathcal{M}}$.
3. Use $\widehat{\mathcal{M}}$ in Algorithm 1 with $N = 100,000$ to obtain an estimate of CASV.

We experiment with $D \in \{100, 1000, 10000\}$ to understand how the finite sample size of data affects the estimate of CASV. In particular, we expect the distribution of the estimator to become more concentrated around the true value as we increase the sample size D .

The results are summarized in Figure 8. As in Figure 4, we condense the state-action attributions into attributions to only actions by aggregating over states *after* computing state-action attributions. Note that the dotted red line in each subfigure of Figure 8 is the attribution corresponding to the true model $\mathcal{M}$, i.e., it corresponds to the attribution displayed under CASV in Figure 4. For instance, the dotted red line in the no-ad column corresponds to an attribution of 50%, which matches the attribution to no-ad under CASV in Figure 4. The histogram in each subfigure characterizes the distribution of the estimator and the solid black line is the empirical mean of the distribution. Each column corresponds to a different ad action and each row corresponds to a different value of D (number of paths used to estimate the Markov model).

Clearly, as we expected, the distribution of the estimator concentrates around the true value as we increase the sample size D . This is perhaps not too surprising since the estimator is asymptotically consistent (because $\widehat{\mathcal{M}} \rightarrow \mathcal{M}$ as $D \rightarrow \infty$). However, it is remarkable that the estimator almost converges to the true value with a value of D as low as 10,000. Contrasting this sample size of 10,000 to the size of our real-world data (several million) highlights the robustness of our numerics presented in Section 6.3.

¹⁸ The notation D should not be confused with the number of Monte Carlo samples N used in Algorithm 1. Instead, D corresponds to the number of user paths we have in our real-world dataset, which is several million.

¹⁹ To be technically correct, we should use the notation $\widehat{\psi}_s^{a,\text{Shap}}(\mathcal{M})$ and $\widehat{\psi}_s^{a,\text{Shap}}(\widehat{\mathcal{M}})$ to acknowledge the Monte Carlo error in Algorithm 1 but as discussed earlier, we found the Monte Carlo error to be negligible with $N = 100,000$.

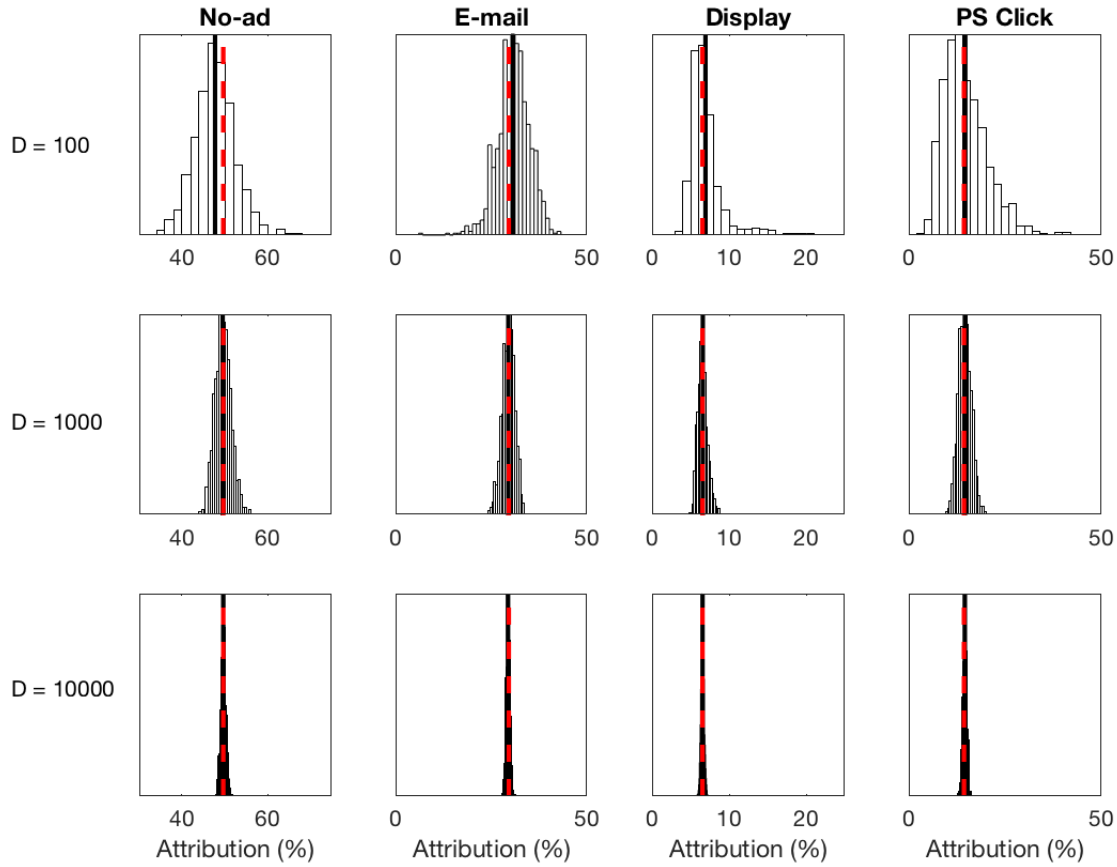


Figure 8 Summary of the sensitivity experiment. Each column corresponds to an ad action and each row corresponds to a different value of D . The dotted red line denotes the true CASV and the histogram denotes the distribution of the estimator. The solid black line equals the empirical mean of the distribution. (We report the percentage attributions to each action by aggregating over states, i.e., $\sum_s \psi_s^a / \sum_{(s', a')} \psi_{s'}^{a'}$ for each $a \in \mathbb{A}$.)

Furthermore, the difference between the solid black line (empirical mean) and the dotted red line (true value) in Figure 8 for $D = 100$ suggests that CASV estimator might be biased. In fact, we believe that to be quite likely. Even if $\widehat{\mathcal{M}}$ is an unbiased estimator of $\mathcal{M}$, it is not necessarily the case that $\psi_s^{a, \text{Shap}}(\widehat{\mathcal{M}})$ is an unbiased estimator of $\psi_s^{a, \text{Shap}}(\mathcal{M})$ since the mapping from $\mathcal{M}$ to $\psi_s^{a, \text{Shap}}$ is non-linear for an arbitrary $(s, a) \in \mathbb{S} \times \mathbb{A}$. Furthermore, characterizing such bias analytically is challenging due to the complicated nature of this mapping. Fortunately, given the enormous size of our real-world dataset (which is quite common in the context of online advertising where data is relatively cheap to collect), we do not have to be concerned about this bias.

Appendix D: Sensitivity of various attributions with respect to τ

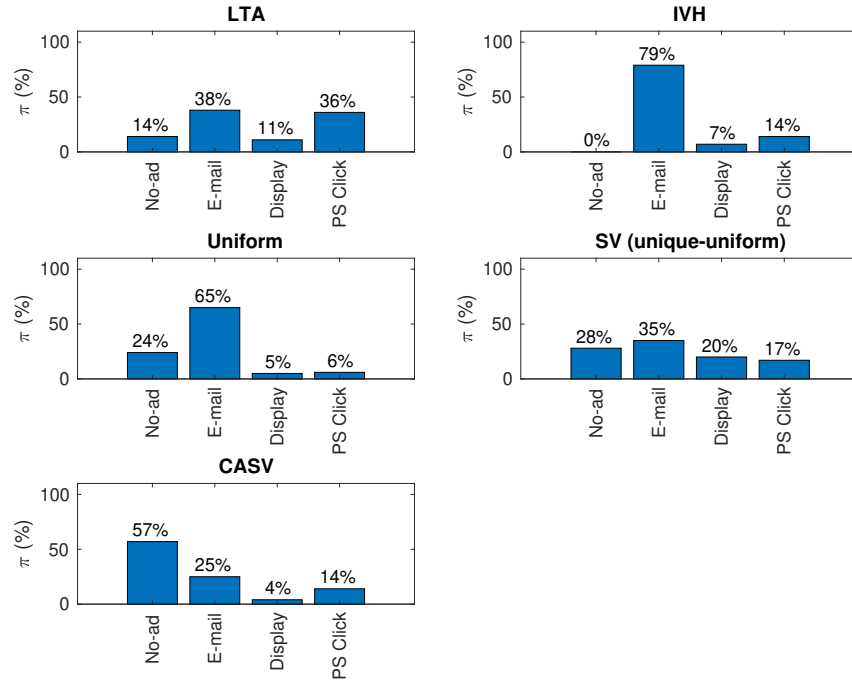


Figure 9 Attributions to different actions under various schemes with $\tau = 7$. We report the percentage attributions to each action by aggregating over states, i.e., $\sum_s \pi_s^a / \sum_{(s', a')} \pi_{s'}^{a'}$ for each $a \in \mathbb{A}$.

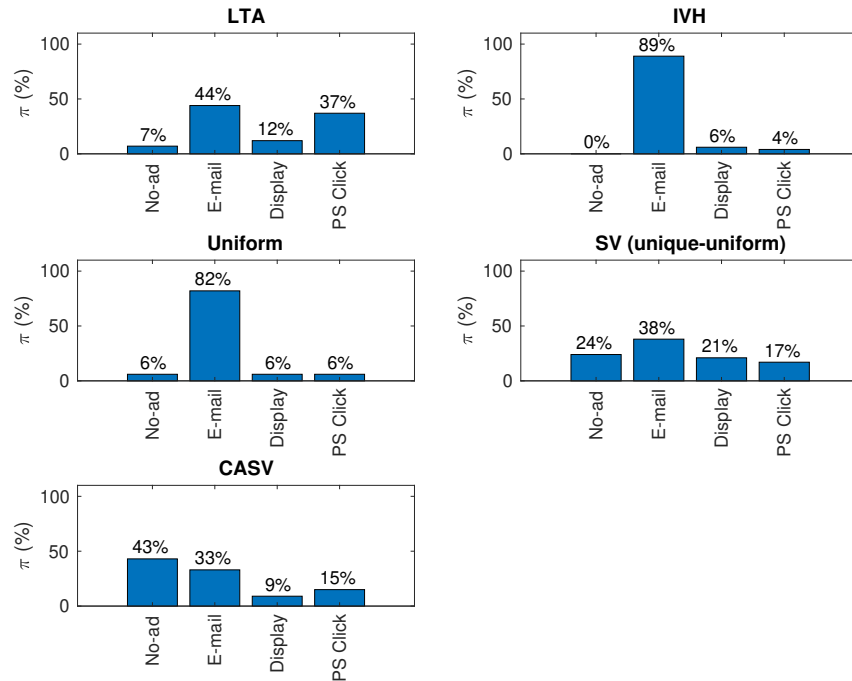


Figure 10 Attributions to different actions under various schemes with $\tau = 14$. We report the percentage attributions to each action by aggregating over states, i.e., $\sum_s \pi_s^a / \sum_{(s', a')} \pi_{s'}^{a'}$ for each $a \in \mathbb{A}$.

Appendix E: Proofs of Theorem 3 and Proposition 2

Theorem 3 (Value of state-specific attribution). *For any action, computing state-specific CASV and then aggregating over states is not the same as computing CASV over the aggregated model. In particular, for all $a \in \mathbb{A}$,*

$$\sum_{s \in \mathbb{S}} \psi_s^{a, \text{Shap}} - \bar{\psi}^{a, \text{Shap}} = \sum_{\mathcal{X} \in \mathbb{T}_1} \{c^a(\mathcal{X}) - \bar{c}^a(\mathcal{X})\} \{v_{\mathcal{M}}(\mathcal{X}) - v_{\bar{\mathcal{M}}^a}(\mathcal{X})\} + \sum_{\mathcal{X} \in \mathbb{T}_2} \sum_{s \in \mathbb{S}} c_s^a(\mathcal{X}) \{v_{\mathcal{M}}(\mathcal{X}) - v_{\mathcal{M}_{(s,a)}}(\mathcal{X})\},$$

where $c^a(\mathcal{X}) := \sum_{s \in \mathbb{S}} c_s^a(\mathcal{X})$,

$$c_s^a(\mathcal{X}) := \begin{cases} w_{|\mathcal{X}|-1} & \text{if } (s, a) \in \mathcal{X} \\ -w_{|\mathcal{X}|} & \text{otherwise} \end{cases} \quad \text{and} \quad \bar{c}^a(\mathcal{X}) := \begin{cases} \bar{w}_{|\bar{\mathcal{X}}|-1} & \text{if } a \in \bar{\mathcal{X}} \\ -\bar{w}_{|\bar{\mathcal{X}}|} & \text{otherwise.} \end{cases}$$

The notation $\bar{\mathcal{X}}$ denotes the actions in $\mathcal{X}$, i.e., $\bar{\mathcal{X}} := \{a : (s, a) \in \mathcal{X}\}$.

Proof. We split the proof into four parts.

Step 1: We use the fact that CASV equals the difference between two SVs

$$\begin{aligned} \psi_s^{a, \text{Shap}}(\mathcal{M}) &= \pi_s^{a, \text{Shap}}(\mathcal{M}) - \pi_s^{a, \text{Shap}}(\mathcal{M}_{(s,a)}) \quad \forall (s, a) \in \mathbb{S} \times \mathbb{A} \\ \bar{\psi}^{a, \text{Shap}}(\mathcal{M}) &= \bar{\pi}^{a, \text{Shap}}(\mathcal{M}) - \bar{\pi}^{a, \text{Shap}}(\bar{\mathcal{M}}^a) \quad \forall a \in \mathbb{A} \end{aligned}$$

to express the difference $\sum_{s \in \mathbb{S}} \psi_s^{a, \text{Shap}} - \bar{\psi}^{a, \text{Shap}}$ as

$$\underbrace{\left[\sum_{s \in \mathbb{S}} \pi_s^{a, \text{Shap}}(\mathcal{M}) - \bar{\pi}^{a, \text{Shap}}(\mathcal{M}) \right]}_{=: (\diamond)} - \underbrace{\left[\sum_{s \in \mathbb{S}} \pi_s^{a, \text{Shap}}(\mathcal{M}_{(s,a)}) - \bar{\pi}^{a, \text{Shap}}(\bar{\mathcal{M}}^a) \right]}_{=: (\bullet)}.$$

Step 2: To analyze $(\diamond)$, we observe the state-specific and aggregated SVs (see (3) and (10)) equal

$$\begin{aligned} \pi_s^{a, \text{Shap}}(\mathcal{M}) &= \sum_{\mathcal{X} \in \mathbb{T}_1 \cup \mathbb{T}_2} c_s^a(\mathcal{X}) v_{\mathcal{M}}(\mathcal{X}) \\ \bar{\pi}^{a, \text{Shap}}(\mathcal{M}) &= \sum_{\mathcal{X} \in \mathbb{T}_1} \bar{c}^a(\mathcal{X}) v_{\mathcal{M}}(\mathcal{X}), \end{aligned}$$

where $\mathbb{T}_1$ and $\mathbb{T}_2$ are the sets of type 1 and 2 coalitions, respectively. This implies

$$(\diamond) = \sum_{\mathcal{X} \in \mathbb{T}_1} \{c^a(\mathcal{X}) - \bar{c}^a(\mathcal{X})\} v_{\mathcal{M}}(\mathcal{X}) + \sum_{\mathcal{X} \in \mathbb{T}_2} \sum_{s \in \mathbb{S}} c_s^a(\mathcal{X}) v_{\mathcal{M}}(\mathcal{X}).$$

Step 3: We analyze $(\bullet)$ similarly to obtain

$$(\bullet) = \sum_{\mathcal{X} \in \mathbb{T}_1} \{c^a(\mathcal{X}) - \bar{c}^a(\mathcal{X})\} v_{\bar{\mathcal{M}}^a}(\mathcal{X}) + \sum_{\mathcal{X} \in \mathbb{T}_2} \sum_{s \in \mathbb{S}} c_s^a(\mathcal{X}) v_{\mathcal{M}_{(s,a)}}(\mathcal{X}),$$

where we use the fact that if $\mathcal{X} \in \mathbb{T}_1$, then $v_{\mathcal{M}_{(s,a)}}(\mathcal{X}) = v_{\bar{\mathcal{M}}^a}(\mathcal{X})$ for all $(s, a) \in \mathbb{S} \times \mathbb{A}$.

Step 4: Putting steps 1, 2, and 3 together concludes the proof. □

Proposition 2. *The ratio of the aggregated model's attribution to the aggregation of the state-specific model's attribution can be arbitrarily large. Mathematically,*

$$\sup_{\mathcal{M}} \sup_{a \in \mathbb{A}} \frac{\bar{\psi}^{a, \text{Shap}}(\mathcal{M})}{\sum_{s \in \mathbb{S}} \psi_s^{a, \text{Shap}}(\mathcal{M})} = \infty.$$

Proof. It suffices to show the existence of one such instance of $(\mathcal{M}, a)$. Consider the network in Figure 11. In the aggregated model, the two players ($a = 1$ and $a = 2$) are symmetric since a user can not convert if either one is absent. Accordingly, for $a \in \{1, 2\}$, $\bar{\psi}^{a, \text{Shap}} = 1/2$. On the other hand, in the state-specific model, players $(1, 2)$ and $\{(s, 1)\}_{s=2}^m$ are null players and the remaining m players are symmetric and hence, each of them receives an attribution equal to the total value in the system divided by m . Accordingly, $\sum_{s \in \mathbb{S}} \psi_s^{1, \text{Shap}} = 1/m$ and $\sum_{s \in \mathbb{S}} \psi_s^{2, \text{Shap}} = (m-1)/m$. Clearly, $\lim_{m \rightarrow \infty} \bar{\psi}^{1, \text{Shap}} / \sum_{s \in \mathbb{S}} \psi_s^{1, \text{Shap}} = \infty$, which concludes the proof.

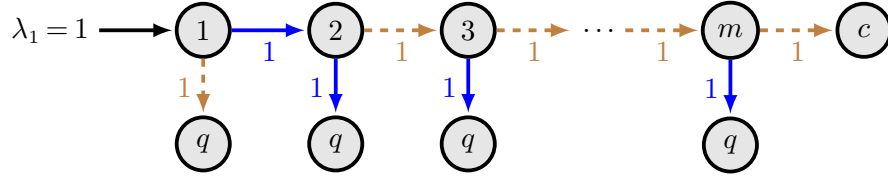


Figure 11 Network for the proof of Proposition 2. The action space consists of three actions: show no-ad, show ad 1, and show ad 2. Solid blue (dashed brown) lines denote transitions if an ad 1 (ad 2) is shown. At state 1, taking action 1 moves the traffic to state 2 w.p. 1 whereas taking action 2 directs the traffic to quit state. At state $s \in \{2, \dots, m\}$, taking action 1 results in a transition to the quit state whereas action 2 moves the users to the “next” state. For brevity, we do not show the transitions if an ad is not shown (to quit state w.p. 1). The advertiser shows ad 1 at state 1 w.p. 1 and ad 2 at all other states w.p. 1.

□